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Exploring Affine Transformations

E. Lee May, Jr.

Department of Mathematics & Computer Science
Salisbury University

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Acknowledgments

This set of notes is the product of the contributions of many. The Educational Advancement Foundation, under the direction of founder Harry Lucas, funded a portion of my sabbatical leave during Academic Year 2004-2005, the period during which I produced the first draft of the notes. The referee of the manuscript that I submitted to the *Journal of Inquiry-Based Learning in Mathematics* (hereafter abbreviated as *JIBLM*), to whom I shall refer as *R*, seemed literally to have gone over the manuscript with a fine-tooth comb, to the point where I believe that *R* now knows the notes as well as I do. In addition, *R* found several errors within the manuscript and made many helpful suggestions for improving it.

The manuscript, like everything that appears in *JIBLM*, was typeset in LaTeX. Although many a reader will respond to this revelation with "Yes. So?", for me to be able to type that sentence is a source of pride, however unwarranted it might be. I had managed, to this point in my career, to avoid learning LaTeX. When my manuscript was accepted for publication in *JIBLM*, however, I knew that the time had come to bite the bullet and learn the language, at least well enough to produce the notes that follow. Upon tackling the challenge, I quickly felt justified at having avoided LaTeXfor so long. How fortunate I was to be able to call on Paul Kapitza for help. During the course of typesetting the manuscript, I have sent him too many e-mails to count. Only one of them did he require more than one iteration and a few hours to answer. Whatever eye-appeal or other attractive quality the manuscript now possesses is due in large part to Paul's guidance and encouragement of me. Whatever flaws remain can be correctly blamed on me.

I am grateful to the students to whom I have taught these notes. The sequence of theorems and exercises is better for the comments that they have made about them. In addition, the enthusiasm with which they tackled the concepts and the success that they experienced in mastering them further encouraged me to publish.

Finally, I gratefully acknowledge the support of my wife, Linda, in the entire process of producing the notes. A mathematician and outstanding software engineer, she has been present throughout the effort. That effort began when I discovered and began to develop some of the fundamental concepts of affine transformations as a graduate student. It has continued for more than forty years and several sabbatical leaves, in addition to time devoted to the topic during many semesters of everyday academic life. As she is in all of my endeavors, so she has been unfailingly supportive, encouraging, and comforting through every step of studying, teaching, and writing about affine transformations. I could not have done it, or much of anything else of significance in my life, without her.

To the Student

This course consists of a study of linear and affine transformations. It has been designed for both graduate and advanced undergraduate students who have successfully completed at least a one-semester course in linear algebra. A purpose of the study is to strengthen each student's understanding of some of the important ideas of linear algebra by extending those ideas to some vector spaces and transformations that usually are not studied in a first course.

The term *transformation*, in modern mathematical language, is another name for *function*. Functions are fundamental elements of mathematics. They are of great interest to mathematicians in and of themselves. They are valuable also, however, for their ability to serve as models of relationships. Functions help us to understand phenomena as diverse as quantum mechanics, planetary motion, and the fluctuations of stock prices.

Linear transformations are the bellwether of all mathematical functions. Not only are they the easiest functions to work with, possessing graphs that are lines or planes through the origin, but also they serve as very accurate models of many physical, biological, and economic phenomena. Life seems to be more nonlinear than linear, however. Affine transformations—most of which, we shall show, are nonlinear—should therefore do even better than linear transformations as models of many phenomena. If, in your academic career, you have studied rigid motion in the plane or computer graphics—or, more likely, if you have seen movies such as *Toy Story*, *Up*, or *Iron Man*—then you have already encountered affine transformations. Affinities, as they are sometimes called, are used heavily also in solving differential equations.

In this study, we shall attempt to discover whether there is any similarity between affine and linear transformations. Put into the form of a question, what properties of linear transformations are possessed also by affinities? Linear transformations on finite-dimensional vector spaces can be represented by matrices. Can something like that be done for affine transformations? Many linear transformations have eigenvalues. Does the term *eigenvalue* make any sense when applied to an affine transformation? Can one go so far even as to use the eigenvalues of an affinity (if they can be shown to exist) to "diagonalize" it, as is done for linear transformations?

In composing "Exploring Affine Transformations," it has been my intention that you learn the topic by a method similar to that employed by professional mathematicians. That method is inquiry. In mathematics, inquiry often assumes the form of formulating and testing conjectures. To verify a conjecture or establish it as true, a mathematician proves a theorem that has that conjecture as its statement. To prove that a conjecture is false, she invents a counterexample to it. Most of the conjectures that appear in this study do so as statements of theorems. It will be your responsibility to convert each such statement from a conjecture that I have labeled as a theorem into a statement that the entire class agrees is a theorem. Your first assignment is to prove Theorem 1 of Chapter 1, and to come to class prepared to present the proof to the rest of the class. (I have assumed, in the preparation of these notes, that most people who use them will be studying affine transformations with others like themselves. For anyone doing an independent study, "class" will probably consist only of a teacher and student.) If, between the first and second meetings of the class, you prove Theorem 1 and have more time that you would like to devote to the study, then try to prove Theorem 2; and so on. At the second meeting of the class, at least one of the people who believe that they have proved Theorem 1 will be asked to present his or her argument. If Theorem 1 is completed and there is time left in the period, someone who has a proof of Theorem 2 will be asked to present that; and so on. If there is time and it seems beneficial for the instructor to say something, he or she will. This is my vision of the format of the meetings of the class.

As you construct your proof of Theorem 1 (and everything that follows it), talk to no one besides your instructor. You may, however, consult not only him or her but also some inanimate resources. One of these is a textbook from an undergraduate course in linear algebra. If you still have the text from the course that you took, that should be more than adequate. If you no longer have access to that book, then any edition of the book, *Linear Algebra and Its Applications*, by David Lay, would make an excellent choice. A second resource is *Finite-Dimensional Vector Spaces*, by the late Paul R. Halmos. A final resource for the course is a textbook or set of notes from a course that often bears a title such as "Introduction to Abstract Mathematics." (A set of such notes is available from me.)

You may cite, in your own proofs, theorems from any of these three types of sources. For example, suppose that, in one of your arguments, you wanted to employ the fact that, if n is a positive integer, then $\mathbf{R}^n$, the space consisting of all ordered n-tuples with addition and scalar multiplication defined componentwise, is a vector space. In such a situation, you could simply say something such as, "By David Lay's statement on Page 217 of *Linear Algebra and Its Applications* (third edition), $\mathbf{R}^n$ is a vector space." You would not need to prove the statement to use it in your argument. For the most part, you need not “re-invent the wheel” in your proofs.

If your meetings with your instructor occur in a classroom setting, I urge you to feel comfortable at any time about volunteering to make a presentation or comment to the rest of the class. Please do this even when you are not sure that you are correct. Behold the turtle: he makes progress only when he sticks his neck out.

These, then, are my recommendations for the study. Thank you for undertaking it. Please feel free to communicate to me any comment that occurs to you about the notes. I believe that you will learn much from playing and working with affine transformations. I believe also that you will experience a satisfaction from your efforts that you have perhaps never known before in mathematics.

E. Lee May, Jr.
elmay@salisbury.edu
Salisbury, Maryland 21801
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Chapter 1

Fundamentals of Affine Transformations, 1

Definition 1. *A* ***linear space*** *is an ordered quadruple (*V, F, +,**) with the following properties:*

(a) V *is a set of one or more points.*

(b) F *denotes either the real field,* R*, or complex field,* C*.*

(c) + is a binary operation on V *and * is a function from* $F \times V$ *into* V *such that the following statements are true:*

(i) if each of ***x*** *and* ***y*** *is in* V*, then* $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$*;*

(ii) if each of ***x***, ***y***, *and* ***w*** *is in* V*, then* $(\mathbf{x} + \mathbf{y}) + \mathbf{w} = \mathbf{x} + (\mathbf{y} + \mathbf{w})$*;*

(iii) there is a point ***0*** *such that* $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$ *for each point* ***x*** *of* V*;*

(iv) for each point ***x*** *of* V *there is a point* ***-x*** *such that* $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$*;*

(v) if c *is a* ***scalar****—that is, a point of* F*—and each of* ***x*** *and* ***y*** *is in* V*, then* $c*(\mathbf{x} + \mathbf{y}) = c*\mathbf{x} + c*\mathbf{y}$*;*

(vi) if each of c *and* d *is in* F *and* ***x*** *is in* V*, then* $(c + d)*\mathbf{x} = c*\mathbf{x} + d*\mathbf{x}$*;*

(vii) if each of c *and* d *is in* F *and* ***x*** *is in* V*, then* $c*(d*\mathbf{x}) = (cd)*\mathbf{x}$*;*

(viii) $1 * \mathbf{x} = \mathbf{x}$ *for each point* ***x*** *of* V.

A linear space is often called a **vector space** or even a **linear vector space**. In addition, the scalar-multiplication operation is often represented as simply juxtaposition—that is, as *c***x** rather than *c*****x**.

Example 2. *Each of the following is a linear space.*

(a) ***R***

(b) ***C***

(c) $\boldsymbol{R}^2$, *often called* ***Euclidean two-dimensional space***

(d) $\boldsymbol{R}^{\mathrm{n}}$, *where* n *is any positive integer (****Euclidean*** n***-dimensional space****)*

(e) *The set of all* 2×2 *matrices of real numbers, with addition and scalar multiplication of matrices defined in the usual way*

Definition 3. *If* ***V*** *denotes the linear space (*V, F, +,**)* *and* ***W*** *denotes the linear space (*W, F, $\oplus$, $\odot$*), then a* ***linear transformation from V to W*** *is a function* T *from* V *to* W *with the following two properties:*

(a) *if each of* ***x*** *and* ***y*** *is a point of* ***V****, then*

$$\mathrm{T}(\boldsymbol{x} + \boldsymbol{y}) = \mathrm{T}\boldsymbol{x} \oplus \mathrm{T}\,\boldsymbol{y}\ ^{1};$$

(b) *if* ***x*** *is a point of* ***V*** *and* c *is in* F, *then*

$$\mathrm{T}(\mathrm{c} * \boldsymbol{x}) = \mathrm{c} \odot \mathrm{T}\boldsymbol{x}.$$

Example 4. *Linear and Nonlinear Transformations*

(a) *The function* f *such that* f(x) = 5x *for each real number* x *is a linear transformation on* ***R****, that is, from* ***R*** *to* ***R****.*

(b) *If* f *is as above, then the function* f + 3*—that is, the function* g *such that* g(x) = f(x) + 3 *for each real number* x*—is not a linear transformation on* ***R****. (*g *is, however, called, at least in most calculus books, a "linear function," presumably because its graph is a line in the plane.)*

(c) *Denote by* T *the function such that* T(x,y) = (x + 2y, 3x + 4y) *for each point* (x,y) *of* $\boldsymbol{R}^2$*.* T *is a linear transformation* ***on*** $\boldsymbol{R}^2$*—that is, from* $\boldsymbol{R}^2$ *to* $\boldsymbol{R}^2$*.*

Exercise 5. *Prove each of the following two statements.*

(a) *If* T *is a linear transformation on* ***R****, then there is a real number* a *such that* Tx = ax *for each number* x.

(b) *If* T *is a linear transformation on* ***C****, then there is a complex number* w *such that* Tz = wz *for each complex number* z.

Notation 6. *Throughout these notes, the letter F denotes either R or C; each of* ***V*** *and* ***W*** *denotes a linear space over* F, *with point-set* V *for* ***V*** *and* W *for* ***W****, and zero-element* $\boldsymbol{0}_{\mathrm{V}}$ *for* ***V*** *and* $\boldsymbol{0}_{\mathrm{W}}$ *for* ***W*** *(when there is no possibility for ambiguity, no subscript will be used);* ***p*** *is a point of* W*—that is, a member of W; and* T *is a linear transformation from* ***V*** *to* ***W****.*

[1] It is customary, in the case of a single argument—say, **x**—to write $T\mathbf{x}$ rather than $T(\mathbf{x})$.

Definition 7. *If* $\boldsymbol{p}$ *is in* W, *then* ***the function constant at*** $\boldsymbol{p}$, *denoted by* $C_{\boldsymbol{p}}$*—or, when there is no possibility for ambiguity, simply by* $\boldsymbol{p}$ *alone—is the function* f *from* V *to* W *such that* f$(\boldsymbol{x}) = \boldsymbol{p}$ *for each point* $\boldsymbol{x}$ *of* V. *The statement, "*f *is a constant-function on* ***V****," means that there is a point* $\boldsymbol{p}$ *of* ***W*** *such that* f $= C_{\boldsymbol{p}}$. *Denote by* Const(V,W) *the set to which a function* f *belongs only in case* f *is a constant-function from* V *to* W, *and by* Const(V) *the set to which* f *belongs only in case* f *is a constant-function from* V *to* V.

Definition 8. *Suppose that* ***V*** *denotes the linear space (*V, F, +,**) and* ***W*** *denotes the linear space (*W, F, $\oplus$, $\odot$*);* M *is a subset of* V*; each of* f *and* g *is a function from* M *into* W*; and* c *is a point of* F. ***The sum of*** f ***and*** g, *denoted* f + g, *is the function from* M *into* W *such that*

$$(f+g)(\boldsymbol{x}) = f(\boldsymbol{x}) + g(\boldsymbol{x})$$

for each point $\boldsymbol{x}$ *of* ***V***. ***The scalar multiple of*** f ***by*** c, *denoted* c*f *or simply* cf, *is the function from* M *into* W *such that*

$$(c*f)(\boldsymbol{x}) = c*f(\boldsymbol{x})$$

for each point $\boldsymbol{x}$ *of* ***V***.

Theorem 9. *(*Const(V,W),F,+,**), where + and * are defined as above, is a linear space.*

Notation 10. *Denote by* ***Const(V,W)*** *the linear space (*Const(V,W),F, +, **), and by* ***Const(V)*** *the linear space (*Const(V),F, +, **).*

Definition 11. *The* ***affine transformation from V to W with base*** T ***and shift*** $\boldsymbol{p}$ *is the transformation* A *such that*

$$A\boldsymbol{x} = T\boldsymbol{x} + \boldsymbol{p}$$

for each point $\boldsymbol{x}$ *of* ***V***. *Such a transformation will sometimes by denoted by* $A_{T,\boldsymbol{p}}$. *The statement, "*A *is* ***strictly affine****," means that* $\boldsymbol{p} \neq \boldsymbol{0}_W$.

Example 12. *In each lettered item below, the function A is a strictly affine transformation.*

(a) $A(x) = x + 1$ *for each real number* x.

(b) $A(x,y) = (x + 2y + 3, 4x + 5y + 6)$ *for each point* (x,y) *of* $\boldsymbol{R}^2$.

(c) *Denote by* T *the linear transformation on* $\boldsymbol{R}^3$ *such that the matrix* $[\mathrm{T}]$ *for* T *with respect to the standard basis*

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}^2$$

is

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

This means that, if $\boldsymbol{x}$ *denotes the point* $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ *of* $\boldsymbol{R}^3$*, then*

$$\mathrm{T}\boldsymbol{x} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} \mathrm{x}_1 \\ \mathrm{x}_2 \\ \mathrm{x}_3 \end{pmatrix} = \begin{pmatrix} \mathrm{x}_1 + 2\mathrm{x}_2 + 3\mathrm{x}_3 \\ 4\mathrm{x}_1 + 5\mathrm{x}_2 + 6\mathrm{x}_3 \\ 7\mathrm{x}_1 + 8\mathrm{x}_2 + 9\mathrm{x}_3 \end{pmatrix}.$$

Denote by $\boldsymbol{p}$ *the point* $\begin{pmatrix} 10 \\ 11 \\ 12 \end{pmatrix}$ *of* $\boldsymbol{R}^3$*, and by* A *the function* $\mathrm{T} + \boldsymbol{p}$.

Exercise 13. *Linear and Strictly-Affine Transformations*

(a) *State the rule for, and draw the graph of, each of three linear transformations from* $\boldsymbol{R}$ *to* $\boldsymbol{R}$.

(b) *State the rule for, and draw the graph of, each of three strictly-affine transformations from* $\boldsymbol{R}$ *to* $\boldsymbol{R}$ *that have not already been presented.*

(c) *State the rule for, and draw the graph of, each of three linear transformations from* $\boldsymbol{R}^2$ *to* $\boldsymbol{R}$.

(d) *State the rule for, and draw the graph of, each of three strictly-affine transformations from* $\boldsymbol{R}^2$ *to* $\boldsymbol{R}$ *that have not already been identified.*

(e) *State the rule for, and draw the graph of, one linear transformation from* $\boldsymbol{R}$ *to* $\boldsymbol{R}^2$ *and one strictly-affine transformation from* $\boldsymbol{R}$ *to* $\boldsymbol{R}^2$.

Notation 14. *Denote by* $\mathrm{Aff}(\boldsymbol{V},\boldsymbol{W})$ *the set consisting of every affine transformation from* $\boldsymbol{V}$ *to* $\boldsymbol{W}$*, and by* $\mathrm{Lin}(\boldsymbol{V},\boldsymbol{W})$ *the subset of* $\mathrm{Aff}(\boldsymbol{V},\boldsymbol{W})$ *consisting of*

[2] In this part of the example, and in any context where a matrix is used to represent the action of a linear transformation, points of the given linear space will be written as column-vectors rather than row-vectors.

every linear member of Aff(***V***,***W***)*. When* W = V, Aff(***V***,***W***) *will be denoted as* Aff(***V***)*, and* Lin(***V***,***W***) *will be denoted as* Lin(***V***)*.*

Definition 15. *Suppose that **V** is the linear space* (V,F,+,*)*;* S *is a subset of* V*; and **S*** = (S,F,+,*)*, where + and * here denote the restrictions to* S *of the addition and scalar-multiplication operations of **V**. Then the statement, "**S** is a **subspace** of **V**," means that all of the following statements are true:*

(a) $\boldsymbol{0}_{\mathrm{V}}$ *is in* S*;*

(b) *if each of **x** and **y** is in* S*, then so is* $\boldsymbol{x} + \boldsymbol{y}$*;*

(c) *if* c *is in* F *and **y** is in* S*, then* c***x*** *is in* S.

*The statement, "**S** is a **proper subspace** of **V**," means that* S *is a proper subset of* V.

Theorem 16. (Aff(***V***,***W***),F,+,*)*, with + as the usual functional addition and * (or juxtaposition) as the usual multiplication of functions by scalars, is a linear space. Moreover, each of* (Lin(***V***,***W***),F,+,*) *and* (Const(***V***,***W***),F,+,*) *is a proper subspace of* (Aff(***V***,***W***),F,+,*)*.*

Notation 17. *With the notation from the preceding theorem, denote by **Aff(V,W)** (**Aff(V)** when* W = V*) the linear space* (Aff(***V***,***W***),F,+,*)*, and by **Lin(V,W)** (**Lin(V)** when* W = V*) the subspace* (Lin(***V***,***W***),F,+,*)*.*

Theorem 18. *Suppose that each of* T_1 *and* T_2 *is a member of **Lin(V,W)** and each of* $\boldsymbol{p}_1$ *and* $\boldsymbol{p}_2$ *is a point of **W** such that*

$$\mathrm{T}_1\boldsymbol{x} + \boldsymbol{p}_1 = \mathrm{T}_2\boldsymbol{x} + \boldsymbol{p}_2$$

*for each point **x** of **V**. Then* $\boldsymbol{p}_1 = \boldsymbol{p}_2$ *and* $\mathrm{T}_1 = \mathrm{T}_2$.

Definition 19. *If* A *is a member of* Aff(***V***,***W***)*, then **the (linear) base of*** A *is the (unique) member T of **Lin(V,W)** with the property that there is a point **p** of* W *such that* A$\boldsymbol{x}$ = T$\boldsymbol{x}$ + $\boldsymbol{p}$ *for each point **x** of* V. T *is sometimes denoted as* T_{A}. *The **shift from*** T ***to*** A *is the (unique) point **p** mentioned immediately above. It might sometimes be denoted as either* $\boldsymbol{p}_{\mathrm{A}}$ *or* $\boldsymbol{p}_{\mathrm{A,T}}$*. In this case, the ordered pair* (T,$\boldsymbol{p}$) *is called **the base-shift pair of*** A.

Definition 20. *Denote by **X** a linear space, and by each of **Y** and **Z** a subspace of **X**. The statement, "**X** is the **sum** of **Y** and **Z**," written as* $\boldsymbol{X} = \boldsymbol{Y} + \boldsymbol{Z}$*, means that*

$$X = Y + Z.$$ [3]

*"$\boldsymbol{X}$ is the **direct sum** of $\boldsymbol{Y}$ and $\boldsymbol{Z}$," written as $\boldsymbol{X} = \boldsymbol{Y} \oplus \boldsymbol{Z}$, means that, if $\boldsymbol{x}$ is in* X*, then there is only one point $\boldsymbol{y}$ of* Y *and there is only one point $\boldsymbol{z}$ of* Z *such that $\boldsymbol{x} = \boldsymbol{y} + \boldsymbol{z}$.*

Example 21. *$\boldsymbol{R}^2 = \boldsymbol{X} \oplus \boldsymbol{Y}$, where $\boldsymbol{X}$ is the linear space with point-set $\{(x,0)$: x is a real number$\}$, and $\boldsymbol{Y}$ is the linear space with point-set $\{(0,y)$: y is a real number$\}$.*

Theorem 22. *If $\boldsymbol{X}$ is a linear space and each of $\boldsymbol{Y}$ and $\boldsymbol{Z}$ is a subspace of $\boldsymbol{X}$, then $\boldsymbol{X} = \boldsymbol{Y} \oplus \boldsymbol{Z}$ only in case*

(a) ***0*** *is the only point of intersection of Y and Z;*

(b) $\boldsymbol{X} = \boldsymbol{Y} + \boldsymbol{Z}$.

Corollary 23. ***Aff(V,W)*** = ***Lin(V,W)*** $\oplus$ ***Const(V,W)***.

Notation 24. *If each of* A *and* B *is a member of* ***Aff(V)***, *denote by* A $\circ$ B, *or simply* AB, *the composition of* A *and* B.

Theorem 25. *If each of* A *and* B *is a member of* ***Aff(V)***, *so is* AB.

Theorem 26. *If each of* A, B, *and* D *is a member of* ***Aff(V)*** *and* Z *is the zero-transformation on* V, *then each of the statements below is true.*

(a) AZ = $C_{\boldsymbol{p}}$, *where* $\boldsymbol{p}$ *is the shift for* A.

(b) ZA = Z.

(c) IA = AI = A, *where* I *denotes the identity transformation on* V.

(d) (A + B)D = AD + BD. *(Additionally, if* D *is linear, then* D(A + B) = DA + DB.*)*

(e) A(BD) = (AB)D.

Corollary 27. *If* A *is an affine transformation on* $\boldsymbol{V}$, *then* AZ = ZA *only in case* A *is linear.*

[3] Note that this is set-equality.

Problem 28. *Is there any connection between the commutativity of affine transformations and that of their linear bases? For example, is the following true?*

> *Suppose that each of* T_1 *and* T_2 *is a member of* $\mathrm{Lin(V)}$*; each of* $\boldsymbol{p}_1$ *and* $\boldsymbol{p}_2$ *is a point of* V*;* $\mathrm{A}_1 = \mathrm{T}_1 + \boldsymbol{p}_1$ *and* $\mathrm{A}_2 = \mathrm{T}_2 + \boldsymbol{p}_2$*; and* $\mathrm{T}_1 \circ \mathrm{T}_2 = \mathrm{T}_2 \circ \mathrm{T}_1$*. Then* $\mathrm{A}_1 \circ \mathrm{A}_2 = \mathrm{A}_2 \circ \mathrm{A}_1$.

Theorem 29. *If* $\mathrm{A} = \mathrm{T} + \boldsymbol{p}$ *and* T *is one-to-one, then so is* A*, and*

$$\mathrm{A}^{-1} = \mathrm{T}^{-1} - \mathrm{T}^{-1}\boldsymbol{p}.$$

Corollary 30. *Under the hypothesis of Theorem 29,* A^{-1} *is affine.*

Theorem 31. *If* A *is a member of* ***Aff(V,W)*** *with base-shift pair* $(\mathrm{T},\boldsymbol{p})$*, then each of the statements below is true.*

(a) A *is one-to-one only in case* T *is.*

(b) A *maps* V *onto* W *only in case* T *does.*

Theorem 32. *Suppose that* S *is a set and each of* f, g, *and* h *is a function from* S *into* S*. Then*

$$\mathrm{f} \circ (\mathrm{g} \circ \mathrm{h}) = (\mathrm{f} \circ \mathrm{g}) \circ \mathrm{h},$$

where $\circ$ *denotes the operation of function-composition.*

Definition 33. *A* ***semigroup*** *is an ordered pair* $(\mathrm{S},\odot)$ *that has the following properties:*

(a) S *is a point-set;*

(b) $\odot$ *is a* ***binary operation on*** S*, that is, a function from* $\mathrm{S} \times \mathrm{S}$ *into* S*;*

(c) $\odot$ *is* ***associative****—that is, if each of* x, y, *and* z *is in* S*, then*

$$\mathrm{x} \odot (\mathrm{y} \odot \mathrm{z}) = (\mathrm{x} \odot \mathrm{y}) \odot \mathrm{z}.$$

Theorem 34. *If* V *is the point-set of a linear space, then* $(\mathrm{Aff(V)},\circ)$ *is a semigroup.*

Definition 35. *A* ***group*** *is an ordered pair* $(\mathrm{S},\odot)$ *that has the following properties:*

(a) S *is a point-set;*

(b) $\odot$ *is an associative binary operation on* S*;*

(c) *there is a point* e *of* S *such that* $x \odot e = x$ *for each point* x *of* S*;*

(d) *for each point* x *of* S*, there is a point* y *of* S *such that* $x \odot y = e$.

Definition 36. *If* S *is a set and* f *is a one-to-one function from* S *into* S*, then the statement, "*f *is invertible," means that* f *is one-to-one and the domain of* f^{-1} *is* S.

Theorem 37. *If* S *is a set and* f *is a one-to-one function from* S *onto* S*, then* f *is invertible.*

Theorem 38. *If* InvAff(V) *denotes the subset of* Aff(V) *to which an affine transformation* A *belongs only in case* A *is invertible, then (*InvAff(V),$\circ$*) is a group.*

Notation 39. *Denote by* ***InvAff(V)*** *the group (*InvAff(V),$\circ$*).*

Chapter 2

Fundamentals of Affine Transformations, 2: The Finite-Dimensional Setting

Exercise 40. *Bases*

(a) *Find a basis, other than the standard one, for* $\boldsymbol{R}^2$.

(b) *Do the same for* $\boldsymbol{R}^3$.

Exercise 41. *In each part of this exercise, find a basis for the given linear space, and use it to determine the dimension of the space.*

(a) *The linear space of all two-by-two matrices*

(b) ***Lin***$(\boldsymbol{R}^2)$*, the linear space of all linear transformations on* $\boldsymbol{R}^2$*—that is, from* $\boldsymbol{R}^2$ *to* $\boldsymbol{R}^2$

(c) *The linear space of all one-by-two matrices*

(d) ***Lin***$(\boldsymbol{R}^2,\boldsymbol{R})$*, that of all linear transformations from* $\boldsymbol{R}^2$ *to* $\boldsymbol{R}$

(e) *The linear space of all three-by-three matrices*

(f) ***Lin***$(\boldsymbol{R}^3)$

Notation 42. *Denote by each of* n *and* m *a positive integer; by* $\boldsymbol{V}$, $\boldsymbol{R}^{\mathrm{n}}$*; and by* $\boldsymbol{W}$, $\boldsymbol{R}^{\mathrm{m}}$*. For each integer-pair* (i,j) *in the Cartesian Product* $[1,\mathrm{m}] \times [1,\mathrm{n}]$*, denote by* $E_{\mathrm{i,j}}$ *the linear transformation from* $\boldsymbol{V}$ *to* $\boldsymbol{W}$ *with matrix*

$$[E_{\mathrm{i,j}}] = \begin{pmatrix} e_{1,1} & \dots & e_{1,n} \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ e_{m,1} & \dots & e_{m,n} \end{pmatrix},$$

where $e_{\mathrm{i,j}} = 1$ *and* $e_{\mathrm{k,l}} = 0$ *if* $(\mathrm{k},\mathrm{l}) \neq (\mathrm{i},\mathrm{j})$.

$[E_{i,j}]$ contains just one nonzero entry, and that entry is a 1 in Row i and Column j. The set

$$\{[E_{i,j}] : i = 1, \ldots, m;\ j = 1, \ldots, n\}$$

is the standard basis for the linear space of all m-by-n matrices, and the set

$$\{E_{i,j} : i = 1, \ldots, m; j = 1, \ldots, n\}$$

is the standard basis for **Lin**(**V**,**W**).

Notation 43. *If* n, m, ***V****, and* ***W*** *are as above and* i *is an integer in [1,*m*], denote by* C_{i} *the constant-function from* ***V*** *to* ***W*** *which is such that, if* ***x*** *is in* V*, then*

$$C_{\mathrm{i}}(\boldsymbol{x}) = (y_1, y_2, \ldots, y_{\mathrm{m}}),$$

where $y_{\mathrm{i}} = 1$ *and* $y_{\mathrm{j}} = 0$ *for each integer* j *in [1,*m*] -* {i}.

Lemma 44. $\{C_1, \ldots, C_{\mathrm{m}}\}$ *is a basis for* ***Const(V)****, which therefore has dimension* m.

Theorem 45. $\{E_{1,1}, \ldots, E_{1,\mathrm{n}}, E_{\mathrm{m},1}, \ldots, E_{\mathrm{m},\mathrm{n}}, C_1, \ldots, C_{\mathrm{m}}\}$ *is a basis for* ***Aff***$(\boldsymbol{R}^{\mathrm{n}}, \boldsymbol{R}^{\mathrm{m}})$.

Corollary 46. ***Aff***$(\boldsymbol{R}^{\mathrm{n}}, \boldsymbol{R}^{\mathrm{m}})$ *has dimension* mn + m*, or* m(n + 1).

Corollary 47. ***Aff***$(\boldsymbol{R}^{\mathrm{n}})$ *has dimension* $\mathrm{n}^2 + \mathrm{n}$*, or* n(n + 1).

Theorem 48. *If* n *is a positive integer,* T *is a member of* Lin$(\boldsymbol{R}^{\mathrm{n}})$*,* ***p*** *is a point of* $\boldsymbol{R}^{\mathrm{n}}$*, and* A = T + ***p****, then the following statements are equivalent:*

(a) A *is one-to-one;*

(b) the ***range of*** A*, which will be denoted* Ran(A)*, is* $\boldsymbol{R}^{\mathrm{n}}$*;*

(c) $\{\boldsymbol{x} : \mathrm{A}\boldsymbol{x} = \boldsymbol{p}\} = \{\boldsymbol{0}\}$.

Chapter 3

“Inflating” an Affine Transformation into a Linear One

Definition 49. *If* n *is a positive integer, then the* ***standard affine*** n-***space****, denoted as* R^n*+1, is the subset of* R^{n+1} *to which the point* $(x_1, x_2, \ldots, x_n, x_{n+1})$ *belongs only in case* $x_{n+1} = 1$*.*

Exercise 50. *Draw a picture of standard affine* n*-space for* n *equal to 1 and then 2. Can you draw or conceptualize standard affine 3-space?*

Notation 51. *If* n *is a positive integer, then* Emb *(for “embed”) denotes the function from* R^n *into* R^n*+1 such that, if* $\boldsymbol{x}$ *denotes the point* $(x_1, x_2, \ldots, x_n)$ *of* R^n*, then*

$$\mathrm{Emb}(\boldsymbol{x}) = (x_1, x_2, \ldots, x_n, 1).$$

Theorem 52. *If* n *is a positive integer, then* Emb *is a one-to-one function from* R^n *onto* R^n*+1.*

Notation 53. Emb$(\boldsymbol{x})$ *will sometimes be written as* Emb$\boldsymbol{x}$.

Theorem 54. *If* n *is a positive integer and* A *is a member of* ***Aff***$(\boldsymbol{R}^n)$*, then there is a member* L *of* ***Lin***$(\boldsymbol{R}^{n+1})$ *with the property that*

$$\mathrm{A}(\boldsymbol{x}) = \mathrm{Emb}^{-1}(\mathrm{L}(\mathrm{Emb}(\boldsymbol{x})))$$

for each point $\boldsymbol{x}$ *of* $\boldsymbol{R}^n$*. (In other words,* L *is such that* $\mathrm{A} = \mathrm{Emb}^{-1} \circ \mathrm{L} \circ \mathrm{Emb}$*.)*

Exercise 55. *When* n *is 3 and then 2, determine what the matrix looks like for the linear transformation described in Theorem 54.*

Corollary 56. *Under the hypothesis of Theorem 54, it is true that*

$$\mathrm{Emb} \circ \mathrm{A} = \mathrm{L} \circ \mathrm{Emb}.$$

Notation 57. *If* n *is a positive integer, then* P_{n+1} *(or sometimes simply* P*) denotes the function from* R^{n+1} *onto* R^n *such that*

$$\mathrm{P}_{n+1}(x_1,x_2,\ldots,x_n,x_{n+1}) = (x_1,x_2,\ldots,x_n)$$

for each point $(x_1,x_2,\ldots,x_n,x_{n+1})$ *of* R^{n+1}.

Theorem 58. *If* n *is a positive integer, then* P_{n+1} *is a linear transformation.*

Definition 59. *Suppose that each of* M, N, *and* S *is a set, with* S *a subset of* M, *and suppose that* f *is a function from* M *into* N. ***The restriction of*** f ***to*** S, *denoted as* $\mathrm{f}\restriction\mathrm{S}$, *is the function from* S *into* N *such that* $(\mathrm{f}\restriction\mathrm{S})(\mathrm{x}) = \mathrm{f}(\mathrm{x})$ *for each point* x *of* S.

Example 60. *Restrictions*

(a) *Suppose that* I^2 *is* ***the square function on*** R*—that is, the function such that* $I^2(\mathrm{x}) = x^2$ *for each real number* x. *Denote by* R^+ *the* ***ray*** $[0,\infty)$. *Then* $I^2 \restriction R^+$ *is the function* f *such that* $\mathrm{f}(\mathrm{x}) = x^2$ *for each number* x *in* R^+.

(b) $P_2 \restriction(\mathrm{R}+1)$ *is the function* g *such that* $\mathrm{g}(\mathrm{x},1) = \mathrm{x}$ *for each point* $(\mathrm{x},1)$ *of* $\mathrm{R}+1$.

Exercise 61. *Produce your own example of a restriction of a function. Draw a two-dimensional picture that illustrates the concept of a restriction of a function.*

Theorem 62. *If* n *is a positive integer, then* $P_{n+1} \restriction (R^n+1) = \mathrm{Emb}^{-1}$.

Theorem 63. *If* n *is a positive integer,* A *is a member of* ***Aff***$(\boldsymbol{R}^n)$, *and* L *is a member of* ***Lin***$(\boldsymbol{R}^{n+1})$ *such that* $\mathrm{A}(\boldsymbol{x}) = \mathrm{Emb}^{-1}(\mathrm{L}(\mathrm{Emb}(\boldsymbol{x})))$ *for each point* $\boldsymbol{x}$ *of* R^n, *then*

$$\mathrm{L}\boldsymbol{u} = (\mathrm{Emb}(\mathrm{A}(\mathrm{P}\boldsymbol{u})))$$

for each point $\boldsymbol{u}$ *of* R^n+1.

Lemma 64. *If* n *is a positive integer and each of* T_1 *and* T_2 *is a linear transformation on* $\boldsymbol{R}^{n+1}$ *such that* $T_1 \restriction (R^n{+}1) = T_2 \restriction (R^n{+}1)$*, then* $T_1 = T_2$.

Theorem 65. *If* n *is a positive integer and* A *is a member of* ***Aff***$(\boldsymbol{R}^n)$*, then there is only one member* L *of* ***Lin***$(\boldsymbol{R}^{n+1})$ *such that*

$$\mathrm{A} = \mathrm{Emb}^{-1} \circ \mathrm{L} \circ \mathrm{Emb}.$$

Definition 66. *If* n *is a positive integer and* A *is a member of* ***Aff***$(\boldsymbol{R}^n)$*, then* ***the inflation of*** A*, denoted* Infl(A)*, is the member of* ***Lin***$(\boldsymbol{R}^{n+1})$ *such that*

$$\mathrm{A} = \mathrm{Emb}^{-1} \circ \mathrm{Infl(A)} \circ \mathrm{Emb}.$$

Exercise 67. *Inflations*

(a) Denote by T *the linear transformation on* $\boldsymbol{R}^2$ *with (standard-basis) matrix*

$$[\mathrm{T}] = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix};$$

by $\boldsymbol{p}$ *the point* $\begin{pmatrix} 5 \\ 6 \end{pmatrix}$ *of* $\boldsymbol{R}^2$*; and by* A *the member* T + $\boldsymbol{p}$ *of* ***Aff***$(\boldsymbol{R}^2)$*. By the preceding theorems,* Infl(A) *is a linear transformation on* $\boldsymbol{R}^3$*, so it has a matricial representation with respect to the standard basis of* $\boldsymbol{R}^3$*. Display that matrix.*

(b) Repeat Exercise a with an example of your own.

(c) Repeat Exercise a with the affine transformation presented in Item (c) of Example 12.

(d) State a conjecture regarding the matricial representation of the inflation of any member ***Aff***$(\boldsymbol{R}^2)$*. Then settle—that is, prove or disprove—your conjecture.*

Suggested Projects

1. Study the set of all two-by-two matrices that have the following form:

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$$

Do they possess any interesting property? For example, is every one of them invertible? Does the set of all of them form a subspace of the linear space of all two-by-two matrices? Does the set of all of them form a sub-semigroup of the semigroup of all two-by-two matrices, with matrix multiplication serving as the semigroup operation? Are they invertible?

2. Repeat Project 1 with the set of all three-by-three matrices of the form

$$\begin{pmatrix} a_{1,1} & a_{1,2} & p_1 \\ a_{2,1} & a_{2,2} & p_2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Theorem 68. *If n is a positive integer and A is a member of* Aff*($\boldsymbol{R}^n$), then*

$$\mathrm{Ran}((\mathrm{Infl(A)} \restriction (R^n + 1)) \subseteqq R^n + 1.$$

Theorem 69. *Under the hypothesis of Theorem 68,*

$$\mathrm{Emb} \circ \mathrm{A} = \mathrm{Infl(A)} \circ \mathrm{Emb}.$$

Definition 70. *If n is a positive integer and A is a member of* Aff*($\boldsymbol{R}^n$) with linear base T, then **the determinant of** T, denoted* $\det_n$*—or, when there is no ambiguity, simply* det*—is the usual determinant of T.*[1]

Exercise 71. *Compute each determinant indicated below.*

(a) That of the linear transformation T such that Tx = πx *for each real number x*

(b) That of the affine transformation A such that Ax = πx + e *for each real number x*

(c) Refer to Example 12 on Page 3. Compute the determinant of the affine transformation A of Item 12(b) .

(d) Again referring to Example 12, compute the determinant of the linear transformation T of Item 12(c) .

Theorem 72. *If n is a positive integer and A is a member of* Aff*($\boldsymbol{R}^n$), then*

$$\det_n(\mathrm{A}) = \det_{n+1}(\mathrm{Infl(A)}).$$

Theorem 73. *If n is a positive integer and A is a member of* Aff*($\boldsymbol{R}^n$), then A is one-to-one only in case* Infl(A) *is.*

Theorem 74. *If n is a positive integer and each of A and B is a member of* Aff*($\boldsymbol{R}^n$), then*

$$\mathrm{Infl(A \circ B)} = \mathrm{Infl(A)} \circ \mathrm{Infl(B)}.$$

Notation 75. *Suppose that S is a set; m is a positive integer; and, for each integer i in [1,m], f_i is a function from S into S. The symbol*

$$\prod_{i=1}^{m} f_i$$

is defined inductively as follows:

[1] This notion of *determinant* can be found in any linear-algebra textbook.

(a) $\prod_{i=1}^{1} f_i = f_1$;

(b) $\prod_{i=1}^{2} f_i = f_1 \circ f_2$;

(c) if j is an integer in [1,m-1] and $\prod_{i=1}^{j} f_i$ *has been defined, then*

$$\prod_{i=1}^{j+1} f_i = (\prod_{i=1}^{j} f_i) \circ f_{j+1}.$$

Corollary 76. *If each of n and m is a positive integer and* A_i *is a member of* Aff*(**R**^{n}) for each integer i in [1,m], then*

$$\prod_{i=1}^{m} \mathrm{Infl}(A_i) = \mathrm{Infl}(\prod_{i=1}^{m} A_i).$$

Theorem 77. *If n is a positive integer, then* Infl *is a one-to-one function* Aff*(**R**^{n}) into* Lin*(**R**$^{n+1}$).*

Notation 78. *Under the hypothesis of Theorem 77, denote* Infl^{-1} *by* Defl. *(*Defl *indicates "deflation.")*

Theorem 79. *If n is a positive integer and each of T and L is a member of* Ran(Infl)*, then*

$$\mathrm{Infl}^{-1}(\mathrm{TL}) = \mathrm{Infl}^{-1}(\mathrm{T}) \circ \mathrm{Infl}^{-1}(\mathrm{L}).$$

An alternate conclusion to Theorem 79 is

$$\mathit{Defl}(TL) = \mathit{Defl}(T) \circ \mathit{Defl}(L).$$

Corollary 80. *If each of n and m is a positive integer and* T_i *is a member of* Ran(Infl) *for each integer i in [1,m], then*

$$\prod_{i=1}^{m} \mathrm{Defl}(T_i) = \mathrm{Defl}(\prod_{i=1}^{m} T_i).$$

Lemma 81. *If n is a positive integer,* I_n *denotes the identity-transformation on* $\boldsymbol{R}^n$*, and* I_{n+1} *denotes that of* $\boldsymbol{R}^{n+1}$*, then*

$$\mathrm{Infl}(\mathrm{I}_n) = I_{n+1}.$$

Theorem 82. *If n is a positive integer and A is a member of* Aff*(**R**^{n}), then A is invertible only in case* Infl(A) *is, and in this case*

$$\mathrm{Infl}(A^{-1}) = (\mathrm{Infl(A)})^{-1}.$$

Chapter 4

Spectral Theory of Affine Transformations, 1: Eigenvalues and Eigenvectors

A study of linear algebra introduces us to the concepts of the eigenvalues and eigenvectors of a linear transformation. One of the lessons of such a study is that these scalars and vectors can be very helpful in performing computations with the linear transformations that possess them. Specifically, we learn that a linear transformation with certain characteristics can be "diagonalized" by means of its eigenvalues and eigenvectors—that is, the matrix of the transformation, with respect to a basis made from its eigenvectors, is diagonal, with its eigenvalues forming the main diagonal.

Computationally, diagonal matrices are the easiest of all matrices with which to work. (If you do not already have a feel for the meaning or truth of this statement, try performing some matrix operations, especially multiplication, with diagonal matrices.) Whenever mathematicians can trade in a non-diagonal matrix for a diagonal one, they leap at the opportunity.

Is there a way to diagonalize affine transformations? Before we began this study, such a question might have seemed ridiculous. We have now seen, however, that affine transformations are not so different from linear ones as their nonlinearity might have led us to believe. For example, the graphs of members of $\mathit{Aff}(\mathbf{R})$ and $\mathit{Aff}(\mathbf{R}^2,\mathbf{R})$ closely resemble those of members of $\mathit{Lin}(\mathbf{R})$ and $\mathit{Lin}(\mathbf{R}^2,\mathbf{R})$, respectively. In addition, although they are not linear, strict affinities can be inflated into (restrictions of) linear transformations. This fact should provide us with some optimism as we, in this chapter, attempt to extend to affine transformations the concepts of *eigenvalue* and *eigenvector*. (We shall postpone until the next chapter our inquiry into diagonalization *per se*.)

Exercise 83. *In a textbook on undergraduate linear algebra, look over the sections dealing with the eigenvalues, eigenvectors, and diagonalization of matrices and linear transformations. Try to (re-)develop a feel for how one decides whether a given linear transformation has an eigenvalue and, when it does, how its eigenvalues and eigenvectors are used to produce a diag-*

onal matrix for it. To test your understanding of linear eigenvalues and eigenvectors, do the following.

(a) *Find a linear transformation on $\boldsymbol{R}^2$—equivalently, a two-by-two matrix—with two eigenvalues.*

(b) *Find a linear transformation on $\boldsymbol{R}^2$ with only one eigenvalue.*

(c) *Find a linear transformation on $\boldsymbol{R}^2$ that fails to have an eigenvalue.*

(d) *Decide whether any of the preceding three linear transformations (matrices) can be diagonalized. If any of them can, try to diagonalize each such transformation.*

(e) *Find a linear transformation on $\boldsymbol{R}^3$—equivalently, a three-by-three matrix—with three eigenvalues.*

(f) *Find a linear transformation on $\boldsymbol{R}^3$ with fewer than three eigenvalues.*

(g) *Decide whether either of the preceding two linear transformations can be diagonalized. If any of them can, try to diagonalize each such transformation.*

(h) *Prove that every linear transformation on $\boldsymbol{R}$ has exactly one eigenvalue.*

Theorem 84. *If T is a linear transformation on $\mathbf{V}$ with eigenvalue λ and associated eigenvector $\boldsymbol{x}$, then there is a point $\boldsymbol{y}$ of $\mathbf{V}$ different from $\boldsymbol{x}$ such that $T\boldsymbol{x} - T\boldsymbol{y} = \lambda(\boldsymbol{x} - \boldsymbol{y})$.*

Corollary 85. *If T is a linear transformation on $\mathbf{V}$ with eigenvalue λ, then there are two points $\boldsymbol{x}$ and $\boldsymbol{y}$ of $\mathbf{V}$ such that $T\boldsymbol{x} - T\boldsymbol{y} = \lambda(\boldsymbol{x} - \boldsymbol{y})$.*

Theorem 86. *Suppose that T is a linear transformation on $\mathbf{V}$ with the property that there is a scalar λ and there are two points $\boldsymbol{x}$ and $\boldsymbol{y}$ of $\mathbf{V}$ such that*

$$\mathrm{T}\boldsymbol{x} - \mathrm{T}\boldsymbol{y} = \lambda(\boldsymbol{x} - \boldsymbol{y}).$$

Then λ is an eigenvalue of T with associated eigenvector $\boldsymbol{x} - \boldsymbol{y}$.

Definition 87. *If A is an affine transformation on $\mathbf{V}$ then the statement, "A has an eigenvalue," means that there is a scalar λ and there are two points $\boldsymbol{x}$ and $\boldsymbol{y}$ of $\mathbf{V}$ such that*

$$\mathrm{A}\boldsymbol{x} - \mathrm{A}\boldsymbol{y} = \lambda(\boldsymbol{x} - \boldsymbol{y}).$$

*In this case λ is called an **eigenvalue of** A; the pair ($\boldsymbol{x}$,$\boldsymbol{y}$) is called an **eigenpair of** (or **for**) A **with respect to** λ; and, if $\mathbf{V}$ is finite-dimensional, **the***

spectrum of *A, denoted $\sigma(A)$, is the set to which the scalar γ belongs only in case γ is an eigenvalue of A.*[1]

Theorems 84 and 86 confirm that this definition specializes to the situation where *A* is linear. In this sense, the new definition can be said to be an **extension** of the definition of *eigenvalue* of a linear transformation. The next theorem further connects the spectrum of an affine transformation with that of its base.

Theorem 88. *If n is a positive integer, T is a linear transformation on $\boldsymbol{R}^n$, and $A = T + \boldsymbol{p}$, then A has an eigenvalue only in case T does; and, in this case, the scalar λ is an eigenvalue of A only case λ is an eigenvalue of T.*

Corollary 89. *If **V** is finite-dimensional and T is a member of **Lin(V)** such that $A = T + \boldsymbol{p}$, then A has a spectrum only in case T does, and in this case $\sigma(A) = \sigma(T)$.*

This says that the eigenvalues of an affine transformation on a finite-dimensional linear space match up exactly with those of it linear base. What is the relationship between eigenvectors and eigenpairs?

Corollary 90. *Under the hypothesis of Theorem 88, the following are true:*

(a) the statement, "$\boldsymbol{x}$ is an eigenvector of T with respect to the eigenvalue λ" implies that $(\boldsymbol{x}, \boldsymbol{0})$ is an eigenpair of A with respect to λ;

(b) "$(\boldsymbol{x},\boldsymbol{y})$ is an eigenpair of A with respect to λ" implies that $\boldsymbol{x} - \boldsymbol{y}$ is an eigenvector of T with respect to λ.

Corollary 91. *Under the hypothesis of Theorem 88, "$(\boldsymbol{x},\boldsymbol{y})$ is an eigenpair of A with respect to λ" implies that $(\boldsymbol{x} - \boldsymbol{y}, \boldsymbol{0})$ is also such a pair.*

Corollary 92. *Under the hypothesis of Theorem 88, $(\boldsymbol{x},\boldsymbol{y})$ is an eigenpair of A with respect to λ only in case $(\boldsymbol{x} - \boldsymbol{y}, \boldsymbol{0})$ is also such a pair.*

Notation 93. *If T is a member of* Lin(***V***) *(**V** is not necessarily finite-dimensional) and λ is an eigenvalue of T, then denote by* Evecs(T,λ) *the subspace of **V** consisting of **0** and every eigenvector of T with respect to λ.*

[1] The spectrum, as it is traditionally defined, in the case where **V** is not finite-dimensional contains not only eigenvalues but also scalars with one of two other properties. Theorem 48 on Page10, however, guarantees that, in the finite-dimensional setting, any scalar that possesses either of these other two properties is also an eigenvalue.

Definition 94. *If each of* $\boldsymbol{V}$ *and* $\boldsymbol{W}$ *is a linear space, then* ***the direct product of V and W****, denoted* $\boldsymbol{V} \times \boldsymbol{W}$*, is the linear space* $(V \times W, F, \oplus, \odot)$ *with the following properties:*

(a) $V \times W$ *is the Cartesian product of* V *and* W*;*

(b) $\oplus$ *is such that, if each of* $\boldsymbol{x}_1 = (\boldsymbol{v}_1, \boldsymbol{w}_1)$ *and* $\boldsymbol{x}_2 = (\boldsymbol{v}_2, \boldsymbol{w}_2)$ *is a point of* $V \times W$*, then* $\boldsymbol{x}_1 \oplus \boldsymbol{x}_2 = (\boldsymbol{v}_1 + \boldsymbol{v}_2, \boldsymbol{w}_1 + \boldsymbol{w}_2)$*;*

(c) $\odot$ *is such that, if* $\boldsymbol{y} = (\boldsymbol{v}, \boldsymbol{w})$ *and* c *is a scalar—that is, a point of* F*—then* $c \odot \boldsymbol{y} = (c\boldsymbol{v}, c\boldsymbol{w})$.

Theorem 95. *The following two statements are true.*

(a) *If each of* $(\boldsymbol{x}, \boldsymbol{y})$ *and* $(\boldsymbol{u}, \boldsymbol{v})$ *is an eigenpair with respect to the eigenvalue* λ *of the affine transformation* A *on* $\boldsymbol{V}$*, then* $(\boldsymbol{x}, \boldsymbol{y}) \oplus (\boldsymbol{u}, \boldsymbol{v})$ *is an eigenpair of* A *with respect to* λ .

(b) *If* $(\boldsymbol{x}, \boldsymbol{y})$ *is an eigenpair with respect to the eigenvalue* λ *of the affine transformation* A *on* $\boldsymbol{V}$ *and* a *is a nonzero scalar, then* $a \odot (\boldsymbol{x}, \boldsymbol{y})$ *is an eigenpair of* A *with respect to* λ.

Corollary 96. *If* A *is an affine transformation on* $\boldsymbol{V}$, λ *is an eigenvalue of* A*, and* $\mathrm{EP}(A, \lambda)$ *denotes the set of all eigenpairs of* A *with respect to* λ*, then* $(\mathrm{EP}(A, \lambda) \cup \{(\boldsymbol{0}, \boldsymbol{0})\}, F, \oplus, \odot)$ *is a subspace of* $\boldsymbol{V} \times \boldsymbol{V}$.

Notation 97. *Denote the subspace of Corollary 96 as* $\boldsymbol{EP(A, \lambda)}$.

Suggested Project. What is the relationship between the eigenvectors and the eigenpairs of a linear transformation? For starters, try a linear transformation of your choice on $\mathbf{R}^2$.

Another question worth asking concerns the eigenvalues of an affine transformation and those of its inflation.

Theorem 98. *If* n *is a positive integer,* A *is a member of* $\mathrm{Aff}(\boldsymbol{R}^n)$ *with linear base* T*, and* T *has eigenvalue* λ *with associated eigenvector* $(x_1, \ldots, x_n)$*, then* λ *is also an eigenvalue of* $\mathrm{Infl(A)}$*, with associated eigenvector* $(x_1, \ldots, x_n, 0)$.

Corollary 99. *If* n *is a positive integer and* A *is a member of* $\mathrm{Aff}(\boldsymbol{R}^n)$ *with eigenvalue* λ *and associated eigenpair* $((x_1, \ldots, x_n), (y_1, \ldots, y_n))$*, then* λ *is also an eigenvalue of* $\mathrm{Infl(A)}$*, with associated eigenvector* $(x_1 - y_1, \ldots, x_n - y_n, 0)$.

Corollary 100. *If* n *is a positive integer and* A *is a member of* $\mathrm{Aff}(\boldsymbol{R}^n)$ *with at least one eigenvalue, then* $\mathrm{Infl(A)}$ *has at least one eigenvalue, and* $\sigma(A) \subseteq \sigma(\mathrm{Infl(A)})$.

Definition 101. *Suppose that F is a function from a subset of the point-set V of the linear space $\boldsymbol{V}$ into V. The statement, " $\boldsymbol{x}$ is a **fixed point of** f," means that $f(\boldsymbol{x}) = \boldsymbol{x}$.*

Theorem 102. *If n is a positive integer, A is a member of $\mathrm{Aff}(\boldsymbol{R}^n)$, and 1 is not an eigenvalue of A, then A has a fixed point.*

Problem 103. *Is the converse of Theorem 102 true?*

Theorem 104. *If n is a positive integer and A is a member of $\mathrm{Aff}(\boldsymbol{R}^n)$, then 1 is an eigenvalue of $\mathrm{Infl(A)}$.*

Corollary 105. *Under the hypothesis of Theorem 104, $\mathrm{Infl(A)}$ has a nonzero fixed point.*

Corollary 106. *If n is a positive integer and A is a member of $\mathrm{Aff}(\boldsymbol{R}^n)$ with at least one eigenvalue, then $\sigma(\mathrm{Infl(A)}) = \sigma(A) \cup \{1\}$.*

Theorem 107. *If n is a positive integer and A is a member of $\mathrm{Aff}(\boldsymbol{R}^n)$, then $\boldsymbol{x}$ is a fixed point of A only in case $\mathrm{Emb}(\boldsymbol{x})$ is a fixed point of $\mathrm{Infl(A)}$.*

Theorem 108. *If n is a positive integer, A is a member of $\mathrm{Aff}(\boldsymbol{R}^n)$, and 1 is not an eigenvalue of A, then there is an eigenvector $(x_1,\ldots,x_n,x_{n+1})$ for $\mathrm{Infl(A)}$ with respect to 1 such that $x_{n+1} = 1$.*

Chapter 5

Spectral Theory of Affine Transformations, 2: Diagonalization

Definition 109. *If n is a positive integer and f is a function from $\boldsymbol{R}^n$ into $\boldsymbol{R}^n$, then the statement,"f is **diagonal**," means that there is a number-sequence $d_1,\ldots,d_n$—called a **defining sequence for** f— such that*

$$f(x_1,\ldots,x_n) = (d_1x_1,\ldots,d_nx_n)$$

for each point $(x_1,\ldots,x_n)$ of $\boldsymbol{R}^n$.

Notation 110. *From now on, diagonal functions will be denoted as D.*

Theorem 111. *If n is a positive integer and D is a diagonal function from $\boldsymbol{R}^n$ into $\boldsymbol{R}^n$, then D is a member of $Lin(\boldsymbol{R}^n)$.*

Theorem 112. *If n is a positive integer and D is a diagonal function from $\boldsymbol{R}^n$ into $\boldsymbol{R}^n$ with defining sequence $d_1,\ldots,d_n$, then*

$$[D] = \begin{pmatrix} d_1 & 0 & 0 & \ldots & 0 \\ 0 & d_2 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 & d_n \end{pmatrix}.$$

A matrix such as [D] is often denoted $diag(d_1,\ldots,d_n)$.

Theorem 113. *If n is a positive integer and D is a diagonal member of $Lin(\boldsymbol{R}^n)$, then* Infl(D) *is a diagonal member of $Lin(\boldsymbol{R}^{n+1})$.*

Definition 114. *Suppose that n is a positive integer and A is a member of* Aff*($\boldsymbol{R}^n$). The statement,"A is **affinely diagonalizable**," means that there is* *an invertible member B of* Aff($\boldsymbol{R}^n$) *and there is a diagonal member D of Lin($\boldsymbol{R}^n$) such that*

$$B^{-1} \circ A \circ B = D.$$

"A is ***linearly diagonalizable****" means that there is an invertible member L of* Lin($\boldsymbol{R}^n$) *and there is a diagonal member D of Lin($\boldsymbol{R}^n$) such that*

$$L^{-1} \circ A \circ L = D.$$

Exercise 115. *Diagonalization*

(a) *Denote by A the affine transformation on $\boldsymbol{R}^2$ such that*

$$[\text{Infl(A)}] = \begin{pmatrix} 10 & 2 & 2 \\ 8 & 4 & 3 \\ 0 & 0 & 1 \end{pmatrix}.$$

Can A be diagonalized? If so, do so. If not, tell why.

(b) *Do the same with the affine transformation A on $\boldsymbol{R}^2$ such that*

$$[\text{Infl(A)}] = \begin{pmatrix} 9 & 7 & 5 \\ 1 & 3 & 7 \\ 0 & 0 & 1 \end{pmatrix}.$$

(c) *Do the same with the affine transformation A on $\boldsymbol{R}^2$ such that*

$$[\text{Infl(A)}] = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(d) *Prove that every member of* Lin($\boldsymbol{R}$) *is linearly diagonalizable. What about every member of* Aff($\boldsymbol{R}$)*?*

Notation 116. *Suppose that S is a set; f is a function from S into S, and n is a nonnegative integer. The symbol f^n is defined inductively as follows:*

(a) *f^0 denotes the identity-transformation on S;*

(b) *f^1 denotes f itself;*

(c) *f^2 denotes $f \circ f$;*

(d) *if* n *is an integer larger than 2 and* f^{n-1} *has been defined, the symbol* f^n *denotes the composition* $f \circ f^{n-1}$.

Theorem 117. *Suppose that* S *is a set, each of* f *and* g *is a function from* S *into* S*, and* h *is an invertible function from* S *into* S*. Then* $hgh^{-1} = f$ *only in case* $h^{-1}fh = g$.

One value of a function's being diagonalizable is illustrated in the next theorem.

Theorem 118. *Suppose that* S *is a set, each of* f *and* g *is a function from* S *into* S*, and* h *is an invertible function from* S *into* S *such that* $hgh^{-1} = f$*. Then*

$$f^n = hg^n h^{-1}$$

for each positive integer n.

As an application of Theorem 118, play with the next exercise.

Exercise 119. [1] *Denote by* L *the linear transformation on* $\boldsymbol{R}^3$ *such that*

$$[L] = \begin{pmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{pmatrix}.$$

(a) *Prove that* L *is diagonalized by the linear transformations* P *and* D*, where*

$$[P] = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

and

$$[D] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

(b) *Compute* $[L]^2$ *by multiplying [L] by itself.*

(c) *Compute* $[L]^2$ *by evaluating* $[P][D]^2[P]^{-1}$.

(d) *Compute* $[L]^3$ *by multiplying [L] by* $[L]^2$.

(e) *Compute* $[L]^3$ *by evaluating* $[P][D]^3[P]^{-1}$.

[1] This is adapted from Example 3, Page 283, of David Lay's *Linear Algebra and Its Applications*, fourth edition.

(f) *Compute $[L]^4$ in two ways, as suggested by what you did to $[L]^3$ in the two immediately-preceding parts of the exercise.*

Theorem 120. *If n is a positive integer and A is a linearly-diagonalizable member of* $\mathrm{Aff}(\boldsymbol{R}^n)$*, then A is linear.*

Corollary 121. *If n is a positive integer and A is a strictly-affine member of* $\mathrm{Aff}(\boldsymbol{R}^n)$*, then A is not linearly diagonalizable.*

Theorem 122. *If n is a positive integer and A is a linearly-diagonalizable member of* $\mathrm{Aff}(\boldsymbol{R}^n)$*, then A is affinely diagonalizable.*

Several questions arise immediately from these definitions and theorems. They will serve as a useful guide to our study in this chapter.

1. Is every affinely-diagonalizable transformation linear? (Another way to ask this is as follows: is there, for some positive integer *n*, a member of $Aff(\mathbf{R}^n)$ - $Lin(\mathbf{R}^n)$ that is affinely diagonalizable?)
2. Is every affinely-diagonalizable transformation linearly diagonalizable?
3. Is there a linear transformation that is not linearly diagonalizable but is affinely diagonalizable?
4. What are some necessary conditions for affine diagonalizability? What are some sufficient conditions for affine diagonalizability?

Theorem 123. *If n is a positive integer and A is an affine-diagonalizable member of* $\mathrm{Aff}(\boldsymbol{R}^n)$*, then* Infl(A) *is a linearly-diagonalizable member of* $\mathrm{Lin}(\boldsymbol{R}^{n+1})$*.*

Corollary 124. ***(Contrapositive)*** *If n is a positive integer and A is a member of* $\mathrm{Aff}(\boldsymbol{R}^n)$ *such that* Infl(A) *is not linearly diagonalizable, then A is not affinely diagonalizable.*

Theorem 125. *Suppose that n is a positive integer; A is a member of* $\mathrm{Aff}(\boldsymbol{R}^n)$ *with base (linear) transformation T; there is an n-element. linearly-independent, set of one or more points of $\boldsymbol{R}^n$, each member of which is an eigenvector of T; and A has a fixed point. Then A is affinely diagonalizable.*

Theorem 126. *Suppose that n is a positive integer; A is a member of* $\mathrm{Aff}(\boldsymbol{R}^n)$ *with base (linear) transformation T; and A is affinely diagonalizable. Then there is a basis for $\boldsymbol{R}^n$ each member of which is an eigenvector of T.*

Corollary 127. *If n is a positive integer and L is an affinely-diagonalizable member of* Lin*($\boldsymbol{R}^n$), then L is linearly diagonalizable.*

Corollary 128. *If n is a positive integer and A is an affinely-diagonalizable member of* Aff*($\boldsymbol{R}^n$) with base T, then T is linearly diagonalizable.*

Corollary 129. *If n is a positive integer and T is a member of* Lin*($\boldsymbol{R}^n$) that is not affinely diagonalizable, then no member of* Aff*($\boldsymbol{R}^n$) that is based on T is affinely diagonalizable.*

Lemma 130. *Suppose that $\boldsymbol{V}$ is a linear space; T is a member of* Lin*($\boldsymbol{V}$); n is a positive integer; $\lambda_1,\ldots,\lambda_n$ is a one-to-one sequence each term of which is an eigenvalue of T; and $\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n$ is a sequence such that $\boldsymbol{x}_i$ is an eigenvector of T with respect to λ_i for each integer i in [1,n]. Then $\{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n\}$ is linearly independent.*

Lemma 131. *If n is a positive integer and A is a member of* Aff*($\boldsymbol{R}^n$) with n eigenvalues, none of which is 1, then* Infl(a) *has n + 1 eigenvalues.*

Lemma 132. *Under the hypothesis of Lemma 131, there is an invertible member B of* Aff*($\boldsymbol{R}^n$) and there is a diagonal transformation D on $\boldsymbol{R}^n$ such that*

$$\mathrm{Infl(B)} \circ \mathrm{Infl(D)} \circ \mathrm{Infl}(B^{-1}) = \mathrm{Infl(A)}.$$

Theorem 133. *If n is a positive integer and A is a member of* Aff*($\boldsymbol{R}^n$) with n eigenvalues, none of which is 1, then A is affinely diagonalizable.*

Corollary 134. *Suppose that n is a positive integer; T is a member of* Lin*($\boldsymbol{R}^n$) that has n eigenvalues, none of which is 1; $\boldsymbol{p}$ is a point of $\boldsymbol{R}^n$; and $A = T + \boldsymbol{p}$. Then A is affinely diagonalizable.*

Conjecture 135. *Each of the following statements is true.*

1. *If n is a positive integer and A is a member of* Aff*($\boldsymbol{R}^n$) with base (linear) transformation T, then A is affinely diagonalizable only in case T is linearly diagonalizable.*
2. *If n is a positive integer and A is a member of* Aff*($\boldsymbol{R}^n$) such that* Infl(A) *is linearly diagonalizable, then A is affinely diagonalizable.*

Chapter 6

Inner-Product Spaces and Adjoints

Exercise 136. *Dot-Products*

(a) *Show that the dot-product $\cdot$ on $\boldsymbol{R}^3$, as it is usually defined in third-semester calculus, has the following properties:*

  (i) *if each of $\boldsymbol{u}$ and $\boldsymbol{v}$ is a point of $\boldsymbol{R}^3$, then $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{u}$;*

  (ii) *if each of $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$ is a point of $\boldsymbol{R}^3$, then*

$$(\boldsymbol{u}+\boldsymbol{w})\cdot\boldsymbol{v} = \boldsymbol{u}\cdot\boldsymbol{v}+\boldsymbol{w}\cdot\boldsymbol{v};$$

  (iii) *if each of $\boldsymbol{u}$ and $\boldsymbol{v}$ is a point of $\boldsymbol{R}^3$ and c is a number, then*

$$(c\boldsymbol{u})\cdot\boldsymbol{v} = c(\boldsymbol{u}\cdot\boldsymbol{v});$$

  (iv) *if $\boldsymbol{u}$ is a point of $\boldsymbol{R}^3$, then $\boldsymbol{u} \cdot \boldsymbol{u} \geqq 0$, and $\boldsymbol{u} \cdot \boldsymbol{u} = 0$ only in case $\boldsymbol{u} = \boldsymbol{0}$.*

(b) *Define the **dot-product on $\boldsymbol{R}^2$** analogously to the dot-product on $\boldsymbol{R}^3$. Show that this dot-product possesses properties analogous to those of the dot-product on $\boldsymbol{R}^3$.*

(c) *Define the **dot-product on $\boldsymbol{R}^4$** analogously to the dot-product on $\boldsymbol{R}^3$. Show that this dot-product possesses properties analogous to those of the dot-product on $\boldsymbol{R}^3$.*

(d) *Denote by n a positive integer, and define the **dot-product on $\boldsymbol{R}^n$** analogously to the dot-product on $\boldsymbol{R}^3$. Show that this dot-product possesses properties analogous to those of the dot-product on $\boldsymbol{R}^3$.*

Definition 137. *If **V** is a linear space over F (with point-set V), then an **inner product on V** is a function, often denoted as $\langle\cdot,\cdot\rangle$, from $V \times V$ into F with the following properties:*

(a) *if each of $\boldsymbol{u}$ and $\boldsymbol{v}$ is a point of **V**—that is, is in V—then $\langle\boldsymbol{u},\boldsymbol{v}\rangle = \langle\boldsymbol{v},\boldsymbol{u}\rangle$;*

(b) *if each of* $\boldsymbol{u}$, $\boldsymbol{v}$, *and* $\boldsymbol{w}$ *is a point of* **V**, *then* $\langle \boldsymbol{u}+\boldsymbol{w},\boldsymbol{v}\rangle = \langle \boldsymbol{u},\boldsymbol{v}\rangle + \langle \boldsymbol{w},\boldsymbol{v}\rangle$;

(c) *if each of* $\boldsymbol{u}$ *and* $\boldsymbol{v}$ *is a point of* V *and* c *is a point of* F, *then* $\langle c\boldsymbol{u},\boldsymbol{v}\rangle = c\langle \boldsymbol{u},\boldsymbol{v}\rangle$;

(d) *if* $\boldsymbol{u}$ *is a point of* V, *then* $\langle \boldsymbol{u},\boldsymbol{u}\rangle \geqq 0$, *and* $\langle \boldsymbol{u},\boldsymbol{u}\rangle = 0$ *only in case* $\boldsymbol{u} = \boldsymbol{0}$.

*An **inner-product space** is an ordered pair* (**V**, $\langle\cdot,\cdot\rangle$) *such that* **V** *is a linear space and* $\langle\cdot,\cdot\rangle$ *is an inner product on* **V**.

Theorem 138. *If each of* $\boldsymbol{u}$, $\boldsymbol{v}$, *and* $\boldsymbol{w}$ *is a point of* **V** *and* c *is a point of* F, *then each of the following is true:*

(a) $\langle \boldsymbol{u},\boldsymbol{v}+\boldsymbol{w}\rangle = \langle \boldsymbol{u},\boldsymbol{v}\rangle + \langle \boldsymbol{u},\boldsymbol{w}\rangle$;

(b) $\langle \boldsymbol{u},c\boldsymbol{v}\rangle = c\langle \boldsymbol{u},\boldsymbol{v}\rangle$;

(c) $\langle \boldsymbol{u}-\boldsymbol{w},\boldsymbol{v}\rangle = \langle \boldsymbol{u},\boldsymbol{v}\rangle - \langle \boldsymbol{w},\boldsymbol{v}\rangle$;

(d) $\langle \boldsymbol{u},\boldsymbol{v}-\boldsymbol{w}\rangle = \langle \boldsymbol{u},\boldsymbol{v}\rangle - \langle \boldsymbol{u},\boldsymbol{w}\rangle$;

(e) $\langle \boldsymbol{0},\boldsymbol{v}\rangle = \langle \boldsymbol{v},\boldsymbol{0}\rangle = 0$.

The concept of *adjoint* is well-known to linear-transformation theorists. If T is a linear transformation on a finite-dimensional linear space **V**, then the (**linear**) **adjoint of** T is that linear transformation T^* with the property that the matrix $[T^*]$ of T^* is the transpose of the matrix $[T]$ of T with respect to a given basis (usually the standard one) of **V**. It is the case that, since every matrix has a transpose, every linear transformation on a finite-dimensional linear space has an adjoint.

Theorem 139. *Suppose that each of* n *and* m *is a positive integer;* T *is a linear transformation from* $\boldsymbol{R}^n$ *to* $\boldsymbol{R}^m$; $\langle\cdot,\cdot\rangle_n$ *is the usual inner product on* $\boldsymbol{R}^n$; *and* $\langle\cdot,\cdot\rangle_m$ *is the usual inner product on* $\boldsymbol{R}^m$. *Then the adjoint* T^* *of* T *is such that* $\langle T\boldsymbol{x},\boldsymbol{y}\rangle_m = \langle \boldsymbol{x},T^*\boldsymbol{y}\rangle_n$ *for each point* $\boldsymbol{x}$ *of* $\boldsymbol{R}^n$ *and each point* $\boldsymbol{y}$ *of* Ran(T).

Theorem 140. *Suppose that each of* n *and* m *is a positive integer;* T *is a linear transformation from* $\boldsymbol{R}^n$ *to* $\boldsymbol{R}^m$; $\langle\cdot,\cdot\rangle_n$ *is the usual inner product on* $\boldsymbol{R}^n$; $\langle\cdot,\cdot\rangle_m$ *is the usual inner product on* $\boldsymbol{R}^m$; *and* L *is a linear transformation from* Ran(T) *to* $\boldsymbol{R}^n$ *with the property that* $\langle T\boldsymbol{x},\boldsymbol{y}\rangle_m = \langle \boldsymbol{x},L\boldsymbol{y}\rangle_n$ *for each point* $\boldsymbol{x}$ *of* $\boldsymbol{R}^n$ *and each point* $\boldsymbol{y}$ *of* Ran(T). *Then* L *is the adjoint of* T*—that is,* $[L]$ *is the transpose of* $[T]$.

Theorems 139 and 140 allow us to make the definition of *adjoint* more general, that is, to extend it to non-finite-dimensional linear spaces.

Definition 141. *Suppose that* $\boldsymbol{H}_1$ *is an inner-product space with inner product* $\langle\cdot,\cdot\rangle_1$*;* $\boldsymbol{H}_2$ *is an inner-product space with inner product* $\langle\cdot,\cdot\rangle_2$*; and* T *is a linear transformation from* $\boldsymbol{H}_1$ *to* $\boldsymbol{H}_2$*. The statement, "*T *has an* ***adjoint****," means that there is a transformation* T^* *from* Ran(T) *to* $\boldsymbol{H}_1$ *with the property that* $\langle T\boldsymbol{x},\boldsymbol{y}\rangle_2 = \langle \boldsymbol{x}, T^*\boldsymbol{y}\rangle_1$ *for each point* $\boldsymbol{x}$ *of* $\boldsymbol{H}_1$ *and each point* $\boldsymbol{y}$ *of* Ran(T).

Can the concept of *adjoint* be extended to affine transformations? Let us see. There are at least two ways in which we might attempt an extension. We could make a first stab by simply replacing the term *linear* with *affine* in Definition 141. The next theorem, however, reveals that, by so doing, we would gain nothing new.

Theorem 142. *Suppose that* $\boldsymbol{H}_1$ *is an inner-product space with inner product* $\langle\cdot,\cdot\rangle_1$*;* $\boldsymbol{H}_2$ *is an inner-product space with inner product* $\langle\cdot,\cdot\rangle_2$*; and* A *is a transformation (not necessarily linear) from* $\boldsymbol{H}_1$ *to* $\boldsymbol{H}_2$ *for which there is a transformation* A^* *with the following properties:*

(a) *the domain of* A^**, denoted* $\mathrm{Dom}(A^*)$*, is a subspace of* $\boldsymbol{H}_2$*, and the range of* A^* *is a subspace of* $\boldsymbol{H}_1$*;*

(b) $\mathrm{Ran(A)} \subseteqq \mathrm{Dom}(A^*)$*;*

(c) $\langle A\boldsymbol{x},\boldsymbol{y}\rangle_2 = \langle \boldsymbol{x}, A^*\boldsymbol{y}\rangle_1$ *for each point* $\boldsymbol{x}$ *of* $\boldsymbol{H}_1$ *and each point* $\boldsymbol{y}$ *of* $\mathrm{Dom}(A^*)$.

Then A *is linear.*

Theorem 142 says that, if one were to define the adjoint of an affine transformation by Properties (a) through (c) of the hypothesis, then the only affine transformations that would have adjoints are the linear ones. A corollary to this theorem would be that no strictly-affine transformation has an adjoint. It seems reasonable to conclude that it is not useful to define the adjoint of an affine transformation by simply parroting the wording of the linear definition.

Before moving on to our second attempt to extend the notion of adjoint, let us note two fascinating and useful properties of adjoints of linear transformations.

Theorem 143. *Suppose that* $\boldsymbol{H}_1$ *is an inner-product space with inner product* $\langle\cdot,\cdot\rangle_1$*;* $\boldsymbol{H}_2$ *is an inner-product space with inner product* $\langle\cdot,\cdot\rangle_2$*; and* T *is a linear transformation from* $\boldsymbol{H}_1$ *to* $\boldsymbol{H}_2$ *with adjoint* T^**. Then the following statements are true:*

(a) $(T^*)^* = T$*;*

(b) T^* *is linear.*

A second way to define the adjoint of an affine transformation is to do something similar to what was done with regard to the eigenvalues of affine transformations. That is to shift from a one-point to a two-point wording.

Definition 144. *Suppose that $\boldsymbol{H}_1$ is an inner-product space with point-set H_1 and inner product $\langle\cdot,\cdot\rangle_1$; $\boldsymbol{H}_2$ is an inner-product space with inner product $\langle\cdot,\cdot\rangle_2$; and A is a transformation (not necessarily linear) from $\boldsymbol{H}_1$ to $\boldsymbol{H}_2$. The statement, "A has an adjoint," means that there is a transformation B with the following properties:*

(a) *the domain of B, denoted* $\mathrm{Dom}(B)$, *is a subspace of $\boldsymbol{H}_2$, and the range of B is a subset of H_1;*

(b) $\mathrm{Ran(A)} \subseteqq \mathrm{Dom}(B)$;

(c) $\langle A\boldsymbol{x} - A\boldsymbol{y}, \boldsymbol{u} - \boldsymbol{v}\rangle_2 = \langle \boldsymbol{x} - \boldsymbol{y}, B\boldsymbol{u} - B\boldsymbol{v}\rangle_1$ *for each point $(\boldsymbol{x},\boldsymbol{y})$ of $H_1 \times H_1$ and each point $(\boldsymbol{u},\boldsymbol{v})$ of* $\mathrm{Dom}(B) \times \mathrm{Dom}(B)$.

Theorem 145. *If each of $\boldsymbol{H}_1$ and $\boldsymbol{H}_2$is an inner-product space and T is a linear transformation from $\boldsymbol{H}_1$ to $\boldsymbol{H}_2$, then T has an adjoint only in case there exists a transformation T^* with the following properties:*

(a) *the domain of T^*, denoted* $\mathrm{Dom}(T^*)$, *is a subspace of $\boldsymbol{H}_2$, and the range of T^* is a subset of H_1;*

(b) $\mathrm{Ran(T)} \subseteqq \mathrm{Dom}(T^*)$;

(c) $\langle T\boldsymbol{x} - T\boldsymbol{y}, \boldsymbol{u} - \boldsymbol{v}\rangle_2 = \langle \boldsymbol{x} - \boldsymbol{y}, T^*\boldsymbol{u} - T^*\boldsymbol{v}\rangle_1$ *for each point $(\boldsymbol{x},\boldsymbol{y})$ of $H_1 \times H_1$ and each point $(\boldsymbol{u}, \boldsymbol{v})$ of* $\mathrm{Dom}(T^*) \times \mathrm{Dom}(T^*)$.

Theorem 145 says that our extended definition of *adjoint* **specializes** to linear transformations.

Theorem 146. *If each of $\boldsymbol{H}_1$ and $\boldsymbol{H}_2$ is an inner-product space, A is a transformation (not necessarily linear) from $\boldsymbol{H}_1$ to $\boldsymbol{H}_2$ that has an adjoint, and each of B_1 and B_2 is an adjoint of A, then $B_1 = B_2$.*

Theorem 146 says that, if an affine transformation possesses an adjoint, it has only one. That fact makes possible the following notational assignment.

Notation 147. *If each of $\boldsymbol{H}_1$ and $\boldsymbol{H}_2$ is an inner-product space, A is a transformation (not necessarily linear) from $\boldsymbol{H}_1$ to $\boldsymbol{H}_2$ that has an adjoint, then denote that (unique) adjoint as A_*.*

Theorem 148. *If each of $\boldsymbol{H}_1$ and $\boldsymbol{H}_2$ is an inner-product space, A is an affine transformation from $\boldsymbol{H}_1$ to $\boldsymbol{H}_2$ with base transformation T, and T has an adjoint T^*, then A also has T^* as its adjoint—that is, $A_* = T^*$.*

Definition 149. *Suppose that* $\boldsymbol{H}$ *is an inner-product space with inner product* $\langle \cdot, \cdot \rangle$*, and* A *is a transformation (not necessarily linear)* ***on*** $\boldsymbol{H}$*—that is, from* $\boldsymbol{H}$ *to* $\boldsymbol{H}$*. The statement, " A is self-adjoint," means that* A *has an adjoint* A_**, and* $A_* = A$*.*

Self-adjointness is very useful in the study of linear transformations on inner-product spaces. For example, see Pages 135 and following and 156 and following in Halmos's *Finite-Dimensional Vector Spaces*.

Corollary 150. *If* $\boldsymbol{H}$ *is an inner-product space and* A *is a strictly-affine transformation on* $\boldsymbol{H}$*, then* A *fails to be self-adjoint.*

This means that, disappointingly, self-adjointness is a property that is peculiar, at least among affine transformations, to the linear ones.

Chapter 7

Norms, Normed Linear Spaces, and Lipschitzian Affinities

Exercise 151. *Setting the Stage*

(a) *Prove that the absolute-value function on* $\boldsymbol{R}$*, denoted* $|\cdot|$*, has the following properties:*

   (i) $|x| \geq 0$ *for each number x;*

   (ii) *if x is a nonzero number, then* $|x| > 0$*;*

   (iii) *if each of a and x is a number, then* $|ax| = |a||x|$*;*

   (iv) *if each of x and y is a number, then* $|x+y| \leq |x| + |y|$.

(b) *Denote by* $||\cdot||$ *the vector-magnitude function on* $\boldsymbol{R}^2$ *or* $\boldsymbol{R}^3$*, as it is defined in calculus. Prove that* $||\cdot||$ *possesses the following properties:*

   (i) $||\boldsymbol{x}|| \geq 0$ *for each point (or vector)* $\boldsymbol{x}$ *of* $\boldsymbol{R}^n$*, where n is 2 or 3;*

   (ii) *if* $\boldsymbol{x}$ *is a nonzero point, then* $||\boldsymbol{x}|| > 0$*;*

   (iii) *if a is a number and* $\boldsymbol{x}$ *is a point, then* $||a\boldsymbol{x}|| = |a| \cdot ||\boldsymbol{x}||$*;*

   (iv) *if each of* $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *is a point, then* $||\boldsymbol{x}+\boldsymbol{y}|| \leq ||\boldsymbol{x}|| + ||\boldsymbol{y}||$.

Theorem 152. *Suppose that* $\boldsymbol{H}$ *is a real inner-product space—that is, the field F of* $\boldsymbol{H}$ *is R—with inner product* $\langle\cdot,\cdot\rangle$*, and that N is the real-valued function defined as follows: N(x) =* $\langle\boldsymbol{x},\boldsymbol{x}\rangle$ *for each point* $\boldsymbol{x}$ *of* $\boldsymbol{H}$*. Then each of the following is true:*

(a) $N(\boldsymbol{x}) \geq 0$ *for each point* $\boldsymbol{x}$ *of* $\boldsymbol{H}$*;*

(b) *if* $\boldsymbol{x} \neq 0$*, then* $N(\boldsymbol{x}) > 0$*;*

(c) *if a is a number and* $\boldsymbol{x}$ *is a point of* $\boldsymbol{x}$*, then* $N(a\boldsymbol{x}) = |a|N(\boldsymbol{x})$*;*

(d) *if each of* $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *is a point of* $\boldsymbol{x}$*, then* $N(\boldsymbol{x}+\boldsymbol{y}) \leq N(\boldsymbol{x}) + N(\boldsymbol{y})$.

Definition 153. *A **normed linear space** is a linear space* $\mathbf{V}$ *with the property that there is a real-valued function* $||\cdot||$ *defined on the point-set* V *of* $\mathbf{V}$ *such that*

(a) $||\mathbf{x}|| \geq 0$ *for each point* $\mathbf{x}$ *of* V;

(b) if $\mathbf{x} \neq 0$, *then* $||\mathbf{x}|| > 0$;

(c) if a *is a number and* $\mathbf{x}$ *is a point of* V, *then* $||a\mathbf{x}|| = |a|||\mathbf{x}||$;

(d) if each of $\mathbf{x}$ *and* $\mathbf{y}$ *is a point of* V, *then* $||\mathbf{x}+\mathbf{y}|| \leq ||\mathbf{x}|| + ||\mathbf{y}||$.

Theorem 154. *If* n *is a positive integer, then* $\boldsymbol{R}^n$ *has a norm.*

Exercise 155. *Invent a second norm for* $\boldsymbol{R}^2$; *for* $\boldsymbol{R}^3$; *for* $\boldsymbol{R}^n$, *where* n *is any positive integer.*

Theorem 156. *If* $\mathbf{V}$ *is a normed linear space and each of* $\mathbf{x}$ *and* $\mathbf{y}$ *is a point of* $\mathbf{V}$, *then* $||\mathbf{x}|| - ||\mathbf{y}|| \leq ||\mathbf{x}-\mathbf{y}||$.

Theorem 157. *If* $\mathbf{V}$ *is a normed linear space and each of* $\mathbf{x}$ *and* $\mathbf{y}$ *is a point of* $\mathbf{V}$, *then* $|||\mathbf{x}|| - ||\mathbf{y}||| \leq ||\mathbf{x}-\mathbf{y}||$.

Theorem 158. *If* $\mathbf{V}$ *is a normed linear space and* $\mathbf{x}$ *is a point of* $\mathbf{V}$, *then* $\mathbf{x} = \mathbf{0}$ *only in case* $||\mathbf{x}|| < \varepsilon$ *for each positive number* ε.

Definition 159. *Suppose that each of* $\mathbf{V}$ *and* $\mathbf{W}$ *is a normed linear space;* $\mathbf{V}$ *has norm* $||\cdot||$ *and* $\mathbf{W}$ *has norm* $|\cdot|$; f *is a function from a subset* Dom(f) *of* V *to* W; *and* $\mathbf{x}$ *is a point of* Dom(f). *The statement, "*f *is **continuous at*** $\mathbf{x}$*," means that, if* ε *is a positive number, then there is a positive number* δ *with the property that, if* $\mathbf{y}$ *is a point of* V *such that* $||\mathbf{x}-\mathbf{y}|| < \delta$, *then* $|f(\mathbf{x}) - f(\mathbf{y})| < \varepsilon$. *The statement, "*$f$ *is **continuous**," means that, if* $\mathbf{x}$ *is any point of* Dom(f), *then* f *is continuous at* $\mathbf{x}$. *If* M *is a subset of* Dom(f) *and* f *is continuous at each point of* M, *then* f *is said to be **continuous on*** M.

Example 160. *Continuous Functions*

(a) The identity-function, I, *is continuous at each point of* $\boldsymbol{R}$. *Thus,* I *is continuous.*

(b) The function f, *where* f(x) = *2x + 3 for each number x, is continuous.*

*(c) The **square-function** (on* $\boldsymbol{R}$*), denoted* I^2*—that is, the function such that* $I^2(x) = x^2$ *for each number x—is continuous.*

(d) *The constant-function* ***7****—that is, the function g such that* g(x) = *7 for each number x—is continuous.*

(e) *The function h defined below is continuous at every number except 0.*

$$h(x) = \begin{cases} -1, & \text{if } x < 0; \\ 1, & \text{if } x \geq 0. \end{cases}$$

(f) *The member T of* Lin(***R***^{2}) *such that*

$$[T] = \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$$

is continuous (with respect to the usual norm) on ***R***^{2}*.*

(g) *The member A of* Aff(***R***^{2}) *such that*

$$[\mathrm{Infl(A)}] = \begin{pmatrix} 9 & 11 & 12 \\ 13 & 17 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

is continuous on ***R***^{2}*.*

Exercise 161. *Find a function from* ***R***^{2} *to* ***R*** *that fails to be continuous at one or more points of* ***R***^{2}*.*

Definition 162. *Suppose that each of* ***V*** *and* ***W*** *is a normed linear space;* ***V*** *has norm* $||\cdot||$ *and* ***W*** *has norm* $|\cdot|$*; f is a function from a subset* Dom(f) *of V to W; and M is a subset of* Dom(f)*. The statement, "f is* ***Lipschitzian on*** *M," means that there is a number b with the property that, if each of* $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *is a point of M, then* $|f(\boldsymbol{x}) - f(\boldsymbol{y})| \leq b||\boldsymbol{x} - \boldsymbol{y}||$*. The statement, "f is* ***Lipschitzian****," means that f is Lipschitzian on* Dom(f)*.*

Example 163. *Lipschitzian Functions*

(a) *The identity-function is Lipschitzian (on* ***R****).*

(b) *The square-function is Lipschitzian.*

(c) *If n is a positive integer, then the identity-function on* ***R***^{n} *is Lipschitzian.*

(d) *The constant-function* ***7*** *is Lipschitzian.*

(e) *The affinity of Example 160(g) is Lipschitzian.*

Exercise 164. *Functions that Fail to be Lipschitzian*

(a) *Find a function f from $\boldsymbol{R}$ to $\boldsymbol{R}$ and a subset M of R such that f fails to be Lipschitzian on M.*

(b) *Do the same with a function from $\boldsymbol{R}^2$ to $\boldsymbol{R}$ and a subset of R^2.*

Theorem 165. *Suppose that **V** is a normed linear space with norm $||\cdot||$, **W** is a normed linear space with norm $|\cdot|$, and T is a member of* Lin*(**V**,**W**). Then T is Lipschitzian only in case there is a number b with the property that, if $\boldsymbol{x}$ is a point of **V**, then $|T\boldsymbol{x}| \leq b||\boldsymbol{x}||$. (In such a case, T is said to be a **bounded linear transformation from V to W**.)*

Theorem 166. *Suppose that each of **V** and **W** is a normed linear space; **V** has norm $||\cdot||$ and **W** has norm $|\cdot|$; f is a function from a subset* Dom(f) *of V to W; and M is a subset of* Dom(f) *on which f is Lipschitzian. Then f is continuous on M.*

Theorem 167. *If n is a positive integer, then every member of* Aff*($\boldsymbol{R}^n$) is Lipschitzian.*

Theorem 168. *Suppose that **V** is a normed linear space with norm $||\cdot||$, **W** is a normed linear space with norm $|\cdot|$, and A is a Lipschitzian member of* Aff*(**V**,**W**). Then there is a smallest nonnegative number b with the property that, if each of $\boldsymbol{x}$ and $\boldsymbol{y}$ is a point of M, then $|A\boldsymbol{x} - A\boldsymbol{y}| \leq b||\boldsymbol{x} - \boldsymbol{y}||$.*

Definition 169. *If **V** is a normed linear space with norm $||\cdot||$, **W** is a normed linear space with norm $|\cdot|$, and A is a Lipschitzian member of* Aff*(**V**,**W**), then the smallest nonnegative number b with the property that, if each of $\boldsymbol{x}$ and $\boldsymbol{y}$ is a point of M, then $|A\boldsymbol{x} - A\boldsymbol{y}| \leq b||\boldsymbol{x} - \boldsymbol{y}||$, is called **the Lipschitz norm** or **Lipschitz bound of** A, and is denoted as $|A|$.*

Theorem 170. *Suppose that **V** is a normed linear space with norm $||\cdot||$, **W** is a normed linear space with norm $|\cdot|$, and* Lip*(**V**,**W**) denotes the set to which A belongs only in case A is a Lipschitzian member of* Aff*(**V**,**W**). Then (*Lip*(**V**,**W**),+,*) (hereafter denoted as **Lip(V, W)**, where + denotes the usual addition of functions and * denotes the usual scalar-multiplication of functions) is a linear space. Moreover, $|\cdot|$, where $|A|$ denotes the Lipschitz bound of A for each member A of* Lip*(**V**,**W**), is a norm on **Lip(V,W)**. (Thus, **Lip(V,W)** is a normed linear space.)*

Theorem 171. *Suppose that **V** is a normed linear space with norm $||\cdot||$, **W** is a normed linear space with norm $|\cdot|$, $\boldsymbol{p}$ is a point of W, T is a member of* Lin*(**V**,**W**), and A is the member $T + \boldsymbol{p}$ of* Aff*(**V**,**W**). Then A is Lipschitzian only in case T is bounded, and in this case $|A| = |T|$.*

Theorem 172. *Suppose that* $\boldsymbol{V}$ *is a normed linear space with norm* $||\cdot||$, $\boldsymbol{W}$ *is a normed linear space with norm* $|\cdot|$, $\boldsymbol{p}$ *is a point of* W, T *is a member of* $\mathrm{Lin}(\boldsymbol{V},\boldsymbol{W})$, *and* A *is the one-to-one member* $T+\boldsymbol{p}$ *of* $\mathrm{Aff}(\boldsymbol{V},\boldsymbol{W})$. *Then* A^{-1} *is Lipschitzian only in case* T^{-1} *is bounded, and in this case* $|A^{-1}| = |T^{-1}|$.

Corollary 173. *Under the hypothesis of Theorem 172, the statement, "A is one-to-one and Lipschitzian," implies that* $|A^{-1}| = |A|^{-1}$.

Chapter 8

Differentiability

The derivative and the integral are the two most important concepts of calculus. Perhaps the most important use of the derivative is that of providing a "local linear approximation" to a function at a number. Specifically, if f is a real-valued function defined on a set of real numbers containing the number x, then $f'(x)$ provides us with a "linear function"—actually, an affine transformation—that approximates f well near the point $(x,f(x))$. With this idea in mind, in this brief chapter we shall extend the notion of *derivative* to functions on normed linear spaces.

Definition 174. *Suppose that each of* $\boldsymbol{V}$ *and* $\boldsymbol{W}$ *is a normed linear space;* $\boldsymbol{V}$ *has norm* $||\cdot||$ *and* $\boldsymbol{W}$ *has norm* $|\cdot|$*;* f *is a function from a subset* Dom(f) *of* V *to* W*; and* $\boldsymbol{x}$ *is a point of* Dom(f)*. The statement, "*f *is* ***differentiable at*** $\boldsymbol{x}$*," means that there is a bounded linear transformation* T *from* $\boldsymbol{V}$ *to* $\boldsymbol{W}$ *with the property that, if* ε *is a positive number, then there is a positive number* δ *such that, if* $\boldsymbol{y}$ *is a point of* V *with the property that* $0 < ||\boldsymbol{x}-\boldsymbol{y}|| < \delta$*, then*

$$\frac{|f(x) - f(y) - T(x-y)|}{||x-y||} < \varepsilon.$$

If S *is a subset of the point-set* V *of* $\boldsymbol{V}$*, then the statement, "*f *is* ***differentiable on*** S*," means that, if* $\boldsymbol{x}$ *is any point of* S*, then* f *is differentiable at* $\boldsymbol{x}$*. If* $S =$ Dom(f)*, then we say that* f *is* ***differentiable****.*

Exercise 175. *Differentiability (and the Lack Thereof)*

(a) *Prove that* I^2 *(see Example 160, which begins on Page 32, Part (c) , for a definition) is differentiable on* $\boldsymbol{R}$*.*

(b) *Do the same with the function 2I + 3—that is, the function* f *such that* f(x) $= 2x + 3$ *for each number* x*.*

(c) *Do the same with the function* I^3*, which is defined analogously to* I^2*.*

(d) *Do the same with E, the function such that* $E(x) = e^x$ *for each number x.*

(e) *Do the same with L, the function such that* $L(x) = ln(x)$ *for each positive number x.*

(f) *Find an example of a differentiable function from* $\boldsymbol{R}^2$ *to* $\boldsymbol{R}$*; from* $\boldsymbol{R}^3$ *to* $\boldsymbol{R}$*.*

(g) *Find an example of a differentiable function from* $\boldsymbol{R}^2$ *to* $\boldsymbol{R}^2$*.*

(h) *Find an example of a function from* $\boldsymbol{R}^2$ *to* $\boldsymbol{R}$ *that fails to be differentiable at one or more points.*

Theorem 176. *Suppose that* $\boldsymbol{V}$ *is a normed linear space with norm* $||\cdot||$*, and* $\boldsymbol{W}$ *is a normed linear space with norm* $|\cdot|$*;* f *is a function from a subset* Dom(f) *of* V *to* W*;* $\boldsymbol{x}$ *is a point of* Dom(f) *at which* f *is differentiable; and each of* T_1 *and* T_2 *is a derivative of* f *at* $\boldsymbol{x}$*. Then* $T_1 = T_2$*.*

In other words, if a function has a derivative at a point, then it has only one derivative there. This fact makes possible the definition that follows.

Definition 177. *Suppose that* $\boldsymbol{V}$ *is a normed linear space with norm* $||\cdot||$*, and* $\boldsymbol{W}$ *is a normed linear space with norm* $|\cdot|$*;* f *is a function from a subset* Dom(f) *of* V *to* W*; and* f *differentiable at the point* $\boldsymbol{x}$ *of* Dom(f)*. Then* ***the derivative of*** f *at* $\boldsymbol{x}$*, which is denoted* f'*(**x**), is the (unique) linear transformation described in Theorem 176.*

Exercise 178. *Prove the following: if* f *is a function from* $\boldsymbol{R}$ *to* $\boldsymbol{R}$ *and* f *is differentiable in "the usual (calculus) sense" at the number x, then* f *is differentiable at x by the definition above. In this case, what is* f'(x)*?*

Theorem 179. *Suppose that* $\boldsymbol{V}$ *is a normed linear space with norm* $||\cdot||$*,* $\boldsymbol{W}$ *is a normed linear space with norm* $|\cdot|$*, and* T *is a bounded linear transformation from* $\boldsymbol{V}$ *to* $\boldsymbol{W}$*. Then* T *is differentiable, and* $T'(\boldsymbol{x}) = T$ *for each point* $\boldsymbol{x}$ *of* $\boldsymbol{V}$*.*

Theorem 180. *Suppose that* $\boldsymbol{V}$ *is a normed linear space with norm* $||\cdot||$*;* $\boldsymbol{W}$ *is a normed linear space with norm* $|\cdot|$*;* T *is a bounded linear transformation from* $\boldsymbol{V}$ *to* $\boldsymbol{W}$*;* $\boldsymbol{p}$ *is a point of* $\boldsymbol{W}$*; and A is the member* $T + \boldsymbol{p}$ *of* ***Aff(V,W)****. Then A is differentiable; and, in fact,* $A'(\boldsymbol{x}) = T$ *for each point* $\boldsymbol{x}$ *of* $\boldsymbol{V}$*.*

What we can now say is that every affine transformation that is based on a bounded linear transformation is differentiable, with its derivative at any point being that base.

Chapter 9

Affine Transformations and Geometry, 1: Motion in the Plane

The simplest form of motion is that along a line. More fascinating is motion in three dimensions, but it is significantly more complicated. In between these two is motion in a plane. Here, what is studied most frequently and profitably is so-called “rigid motion.” There are three types of rigid motion: rotation about a point, reflection in a line, and translation. One type of non-rigid motion at which we shall also take a look is scaling, that is, shrinking or stretching.

9.1 Rotation About the Origin

Theorem 181. *Suppose that each of λ, p_1, and p_2 is a number, and $\{p_1,p_2\} \neq \{0\}$; T is the member λI of $\mathrm{Lin}(\boldsymbol{R}^2)$; and A is the affine transformation $T + (p_1,p_2)$. Then A is affinely diagonalizable only in case $\lambda \neq 1$, and in this case $A = BDB^{-1}$, where*

$$[\mathrm{Infl(B)}] = \begin{pmatrix} 1 & 0 & p_1/(1-\lambda) \\ 0 & 1 & p_2/(1-\lambda) \\ 0 & 0 & 1 \end{pmatrix}$$

and $D = \lambda I$.

Definition 182. *If θ is a number in the half-open interval $[0,2\pi)$ and T is the linear transformation on $\boldsymbol{R}^2$ with (standard-basis) matrix*

$$[T] = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

then T is ***the rigid motion of rotating, in a counterclockwise direction, every vector*** [1] ***through an angle of θ about the origin****.*

[1] In this chapter, it might be helpful to interpret *vector*—or equivalently, *point*—as a directed line segment

Exercise 183. *Produce some examples that provide evidence that this definition is reasonable.*

Theorem 184. *If θ is a number in [0,2π) and T rotates every vector through an angle of θ about the origin, then T has an eigenvalue only in case θ is 0 or π/2. In the former case, T = I, the identity-transformation on $\boldsymbol{R}^2$; in the latter, T = -I. In both cases, T is diagonal, hence* a fortiori *linearly diagonalizable. Moreover, no strictly-affine transformation based on I is affinely diagonalizable, but every one based on -I is.*

Corollary 185. *The following statement is false: if T is a linearly-diagonalizable linear transformation on the linear space **V**, **p** is a point of **V**, and A = T + **p**, then A is affinely diagonalizable.*

9.2 Rotation About a Point Other Than the Origin

Theorem 186. *If each of a and b is a number, at least one of which is nonzero, and θ is a number in [0,2π), then the **rigid motion of rotating every point of $\boldsymbol{R}^2$ counterclockwise through an angle of** θ **about** (a,b) is an affine transformation from $\boldsymbol{R}^2$ to $\boldsymbol{R}^2$.*

Exercise 187. *Denote by A the affinity mentioned in the conclusion of Theorem 186. Display [*Infl(A)*].*

Note that the affinity just defined is a special case of the affinities mentioned in Theorem 184, that is, ones formed by rotating every point of $\mathbf{R}^2$ counterclockwise through an angle of θ about the origin and then translating the resulting point.

Theorem 188. *If each of a and b is a number, at least one of which is nonzero; θ is a number in [0,2π); and A is the rigid motion of rotating counterclockwise every point of $\boldsymbol{R}^2$ through an angle of θ about (a,b), then A = BTB^{-1}, where*

$$[Infl(B)] = \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

and

$$[Infl(T)] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

beginning at the origin. As an exercise, draw some pictures for specific angles to verify that describing T in this way is reasonable.

Theorem 189. *If each of a and b is a number, θ is a number in $[0,2\pi)$, and A is the rigid motion of rotating counterclockwise every point of $\mathbf{R}^2$ through an angle of θ about (a,b), then A is diagonalizable only in case θ is 0 or $\pi/2$. In the former case, A = T = I; in the latter,*

$$[Infl(A)] = \begin{pmatrix} \cos\theta & -\sin\theta & a(1-\cos\theta)+b\sin\theta \\ \sin\theta & \cos\theta & b(1-\cos\theta)-a\sin\theta \\ 0 & 0 & 1 \end{pmatrix}.$$

Exercise 190. *It is claimed that, if one wishes to rotate every point of the plane through an angle of θ about a point (a,b) other than the origin, one can just as easily achieve the same effect by rotating every point of the plane through an angle of θ about the origin, and then translating the resulting point by the point*

$$(a(1-\cos\theta)+b\sin\theta, b(1-\cos\theta)-a\sin\theta).$$

Comment on the accuracy—that is, the truthfulness—of this claim.

9.3 Scaling Relative to the Origin

Theorem 191. *If each of s and t a number, then the motion of **scaling**—think "shrinking or stretching"— **every vector of $\mathbf{R}^2$** by (s,t) **relative to the origin** is a diagonal transformation.*

Exercise 192. *Denote by D the transformation mentioned in the conclusion of Theorem 191. Display [D].*

Theorem 193. *Suppose that each of s and t is a number, D is the motion of scaling by (s,t) relative to the origin, and A is a strictly-affine transformation based on D—that is, there is a point (p_1,p_2) of $\mathbf{R}^2$ such that at least one of p_1 and p_2 is different from 0 and $A = D + (p_1, p_2)$. Then the following statements are true.*

(a) If $s = t$, then A is affinely diagonalizable only if s differs from 1.

(b) If $s \neq t$, then

(i) if neither s nor t is 1, then A is affinely diagonalizable;

(ii) if s =1, then A is affinely diagonalizable only in case p_1 = 0;

(iii) if t =1, then A is affinely diagonalizable only in case p_2 = 0.

9.4 Scaling Relative to a Point Other Than the Origin

Theorem 194. *If each of a and b is a number, at least one of which is nonzero, and each of s and t is a number, then the motion of scaling every vector of* $\boldsymbol{R}^2$ *by (s,t) relative to* (a,b) *is an affine transformation.*

Exercise 195. *Denote by A the affinity mentioned in the conclusion of Theorem 194. Display [*Infl(A)*].*

Theorem 196. *If each of a and b is a number, at least one of which is nonzero; each of s and t is a number; and A is the rigid motion of scaling by (s,t) relative to (a,b), then A is affinely diagonalizable unless* $s = t = 1$.

9.5 Reflection in a Line Through the Origin

Theorem 197. *Suppose that each of a and b is a number such that* $a^2 + b^2$ *= 1;* $\boldsymbol{u}$ *= (a,b); and* L *is the line* $\{t\boldsymbol{u} : t \text{ is a number}\}$. *Then* ***the rigid motion of reflecting every vector in*** L *is a linear transformation from* $\boldsymbol{R}^2$ *to* $\boldsymbol{R}^2$.

Note that the line L of Theorem 197 "passes through"—that is, contains—the origin.

Exercise 198. *Denote by T the transformation mentioned in the conclusion of Theorem 197. Display [T].*

Theorem 199. *Suppose that each of a and b is a number such that* $a^2 + b^2$ *= 1;* $\boldsymbol{u}$ *= (a,b);* L *is the line* $\{t\boldsymbol{u} : t \text{ is a number}\}$; *T is the rigid motion of reflecting every vector in* L; $\boldsymbol{p}$ *is the point* (p_1,p_2) *of* $\boldsymbol{R}^2$; *and A = T +* $\boldsymbol{p}$. *Then A is affinely diagonalizable only in case* $ap_1 + bp_2$ *= 0; and in this case* $A = BDB^{-1}$, *where*

$$[\mathrm{Infl(D)}] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and B is such that

$$[\mathrm{Infl(B)}] = \begin{pmatrix} a & -b & p_1/(2b^2) \\ b & a & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

if $b \neq 0$*, and*

$$[\mathrm{Infl(B)}] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & p_2/2 \\ 0 & 0 & 1 \end{pmatrix}$$

otherwise.

9.6 Reflection in a Line Not Through the Origin

Theorem 200. *Suppose that each of a and b is a number such that a^2+b^2 = 1; $\boldsymbol{u}$ = (a,b); each of p and q is a number at least one of which is different from 0; $\boldsymbol{v}$ = (p,q); and L is the line $\{\boldsymbol{x} : \boldsymbol{x} = t\boldsymbol{u} + \boldsymbol{v}$ for some number $t\}$. Then* ***the rigid motion of reflecting every vector in*** *L is an affine transformation from $\boldsymbol{R}^2$ to $\boldsymbol{R}^2$.*

Exercise 201. *Denote by A the affinity mentioned in the conclusion of Theorem 200. Display [*Infl(A)*].*

Theorem 202. *Suppose that each of a and b is a number such that a^2+b^2 = 1; $\boldsymbol{u}$ = (a,b); each of p and q is a number at least one of which is different from 0; $\boldsymbol{v}$ = (p,q); L is the line $\{\boldsymbol{x} : \boldsymbol{x} = t\boldsymbol{u} + \boldsymbol{v}$ for some number $t\}$; and A is the rigid motion of reflecting every vector in L. Then A is affinely diagonalizable. In fact, $A = BDB^{-1}$, where*

$$[\mathrm{Infl(D)}] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and B is such that

$$[\mathrm{Infl(B)}] = \begin{pmatrix} a & -b & (pb-qa)/b \\ b & a & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

if $b \neq 0$; and

$$[\mathrm{Infl(B)}] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & a(aq-bp) \\ 0 & 0 & 1 \end{pmatrix}$$

otherwise.

Chapter 10

Affine Transformations and Geometry, 2: Affine Combinations

Definition 203. *Suppose that n is a positive integer, each of $\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_n$ is a point of the linear space **V**, and **x** is a point of **V**. The statement, "**x** is an **affine combination of** $\{\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_n\}$," means that there is a number-sequence $a_1, a_2, \ldots, a_n$ such that*

$$\sum_{i=1}^{n} a_i = 1$$

and

$$\boldsymbol{x} = \sum_{i=1}^{n} a_i \boldsymbol{y}_i.$$

Theorem 204. *Suppose that each of **V** and **W** is a linear space; A is an affine transformation from **V** to **W**; a is a number; and each of **x** and **y** is a point of **V**. Then*

$$A(a\boldsymbol{x} + (1-a)\boldsymbol{y}) = aA\boldsymbol{x} + (1-a)A\boldsymbol{y}.$$

Definition 205. *Suppose that each of **x** and **y** is a point of the linear space **V**. **The affine span of** $\{x, y\}$, denoted as $\mathrm{AS}(\{\boldsymbol{x},\boldsymbol{y}\})$, is the set*

$$\{a\boldsymbol{x} + (1-a)\boldsymbol{y} : a \text{ is a number}\}.$$

*If $\boldsymbol{x} \neq \boldsymbol{y}$, then $\mathrm{AS}(\{\boldsymbol{x},\boldsymbol{y}\})$ is called **the line through** (or **containing**) **x** and **y**, and is denoted as $\mathrm{Line}\{\boldsymbol{x},\boldsymbol{y}\}$.*

Corollary 206. *If each of **V** and **W** is a linear space, A is an affine transformation from **V** to **W**, and **x** and **y** are two points of **V**, then $A(\mathrm{Line}\{\boldsymbol{x},\boldsymbol{y}\})$ =*

Line$\{A\boldsymbol{x}, A\boldsymbol{y}\}$.

Theorem 207. *Suppose that each of* $\boldsymbol{V}$ *and* $\boldsymbol{W}$ *is a linear space and* A *is a transformation from* $\boldsymbol{V}$ *to* $\boldsymbol{W}$ *with the property that, if* a *is a number and each of* $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *is a point of* $\boldsymbol{V}$, *then*

$$A(a\boldsymbol{x} + (1-a)\boldsymbol{y}) = aA\boldsymbol{x} + (1-a)A\boldsymbol{y}.$$

Then A *is an affine transformation.*

Conjecture 208. *Suppose that each of* $\boldsymbol{V}$ *and* $\boldsymbol{W}$ *is a linear space, and that* A *is a transformation from* $\boldsymbol{V}$ *to* $\boldsymbol{W}$ *with the property that, if* a *is a number and* $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *are two points of* $\boldsymbol{V}$, *then*

$$A(\text{Line}\{\boldsymbol{x}, \boldsymbol{y}\}) = \text{Line}\{A\boldsymbol{x}, A\boldsymbol{y}\}.$$

Then A *is an affine transformation.*

If Conjecture 208 is true, then it can be combined with Corollary 206 to yield the following theorem:

> Suppose that each of **V** and **W** is a linear space and A is a transformation from **V** to **W**. Then a necessary and sufficient condition that A be affine is that, if **x** and **y** are two points of **V**, then A(*Line*$\{$**x**,**y**$\}$) = *Line*$\{$A**x**,A**y**$\}$.

The condition of mapping lines to lines is often used, especially in textbooks on geometry, as the defining property of affine transformations. A proof of the conjecture, and hence of the theorem immediately resulting from it, would show that the mapping of lines to lines is equivalent to the condition that has been used to define *affine transformation* in these notes.

Theorem 209. *Suppose that each of* $\boldsymbol{V}$ *and* $\boldsymbol{W}$ *is a linear space;* A *is an affine transformation from* $\boldsymbol{V}$ *to* $\boldsymbol{W}$*;* n *is a positive integer; each of* $\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_n$ *is a point of* $\boldsymbol{V}$*;* $a_1, a_2, \ldots, a_n$ *is a number-sequence such that*

$$\sum_{i=1}^{n} a_i = 1;$$

and $\boldsymbol{x}$ *is the affine combination*

$$\sum_{i=1}^{n} a_i \boldsymbol{y}_i.$$

Then

$$A\boldsymbol{x} = \sum_{i=1}^{n} a_i A\boldsymbol{y}_i.$$

Thus, just as a linear transformation preserves linear combinations (can you prove this?), so an affine transformation preserves affine combinations. Another way of saying this is that the preservation of affine combinations is a necessary condition for a transformation to be affine.